

Gatekeepers and Self-Preferencing: Incentives and Welfare Trade-offs in Two-sided Markets*

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Abstract

We study a two-sided platform where a dominant gatekeeper supplies both a primary product and a complementary ancillary service. By degrading outcomes for users of rival ancillary services, the gatekeeper engages in cross-market self-preferencing. Our theoretical model identifies when such behavior raises or lowers welfare, depending on buyers' and sellers' preferences and the platform's pricing instruments. Analyzing a recent antitrust case against Google, we show that remedies targeting only one side can misalign incentives between different user groups, reducing welfare. Counterfactual simulations highlight when alternative behavioral or structural interventions realign incentives and improve market efficiency.

Keywords: Multi-Sided Market; Platform Economics; Competition Policy; Digital Advertising.

JEL Codes: L21, L40, L51.

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“For over a decade, Google has tied its publisher ad server and ad exchange together through contractual policies and technological integration, which enabled the company to establish and protect its monopoly power in these two markets. . . . In addition to depriving rivals of the ability to compete, this exclusionary conduct substantially harmed Google’s publisher customers, the competitive process, and, ultimately, consumers of information on the open web.” — Judge Leonie Brinkema, *United States v. Google LLC*, April 17, 2025.

1 Introduction

In two-sided markets, users derive value from interacting with each other and often benefit from a growing user base. The network effects enhance the appeal of dominant companies, resulting in market concentration. In addition to their core services, dominant digital platforms (i.e., *gatekeepers*) have continuously expanded into ancillary service markets (see Athey and Scott Morton (2021), Hagiu et al. (2022), and Heidhues et al. (2024)). These ancillary services often complement the core products offered in the primary market, facilitating transactions between buyers and sellers. For instance, Amazon offers product storage for sellers to fulfill Amazon orders, Apple provides electronic payment technologies or subscription channels for iOS developers to interact with mobile users, and Google supplies multiple complementary technologies to facilitate interactions in the digital advertising market. Although these firms hold dominant market shares in their primary markets, they often face competition in ancillary service markets. However, they can leverage their dominance in the primary market to promote and advantage their own ancillary service—this practice is commonly referred to as *self-preferencing*.

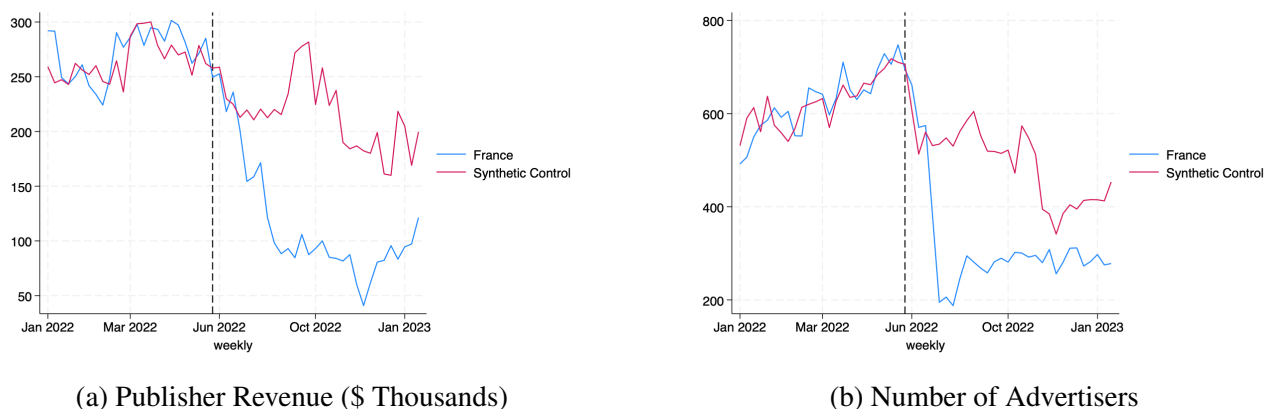
In recent years, self-preferencing has attracted growing regulatory scrutiny. In 2025, the U.S. Department of Justice (DoJ) won a landmark case against Google, ruling that the company had monopolized digital advertising markets by favoring its own advertising technology (adtech) tools. Similar practices have prompted enforcement actions in Europe: Amazon was required by the European Commission (EC) to modify its Buy Box recommendation algorithm to prevent it from favoring merchants using Amazon’s logistics services, and Apple was fined €1.8 billion for deterring app developers from advertising cheaper subscription options outside the App Store. The Digital Markets Act¹ now explicitly prohibits self-preferencing by designated gatekeepers in the European Union (EU), while forthcoming guidelines² on exclusionary abuses from the EC are expected to set obligations for dominant firms to avoid such behavior.

¹See https://digital-markets-act.ec.europa.eu/index_en.

²See https://competition-policy.ec.europa.eu/public-consultations/2024-article-102-guidelines_en.

Despite the regulatory momentum, the economic analysis of self-preferencing—and its broader welfare implications—has remained relatively limited. Moreover, in some cases, the ex post remedies designed to curb such practices have unintentionally exacerbated the very issues they sought to address. A notable example is the French Competition Authority (FCA)’s 2021 antitrust case against Google in the adtech market, a critical and highly complex two-sided market (see institutional background in the Appendix). In short, the FCA examined two complementary and mandatory technology layers used by publishers to interact with advertisers. Google held a dominant position in one of these layers and was found to leverage its dominance to disadvantage rivals in the other. As part of the settlement, Google committed to reducing its self-preferencing practices in order to level the playing field. However, Figure 1 illustrates that, following the June 2022 implementation deadline of commitment, publisher revenue and advertiser participation for major domains in France declined markedly compared with a synthetic control group (This case is analyzed in greater detail in Section 4). These findings highlight the unintended consequences and complications of regulating self-preferencing in digital ecosystems.

Figure 1: Display Advertising Market: Pre- and Post-FCA Intervention



The dashed line marks June 30, 2022, the deadline set by the FCA for implementing Google’s data-sharing commitment.

Motivated by these observations, this paper develops a model of self-preferencing that reflects key market characteristics and helps explain the puzzling evidence emerging from antitrust interventions. The core feature of the model is based on a simple yet powerful insight: the ancillary adtech, used alongside publishers’ primary adtech, is chosen and often paid for by publishers (i.e., sellers). However, these choices are employed to interact with advertisers (i.e., buyers) or their ad-buying tools, thereby influencing their surplus. In other words, both buyers and sellers have preferences over the ancillary services, and these preferences can either align or diverge. The model accounts for this flexibility and the two-sided nature of preferences. Given the complex interplay between primary products and ancillary services, as well as the emergence of novel market features, the underlying mechanisms and

trade-offs captured by our model differ significantly from those in existing research, as discussed later.

Specifically, the benchmark model features two user groups—buyers and sellers—who derive value from interacting with each other and have heterogeneous outside options. To enable these interactions, sellers must rely on a primary product supplied by a dominant firm. In addition, they also have to purchase an ancillary service in an adjacent market to complete transactions with buyers. While the primary product is monopolized, ancillary services are available both from the dominant firm (*gatekeeper*) and from independent providers. We define *cross-market self-preferencing* as the practice where the gatekeeper promotes its own ancillary services by reducing the relative profitability of sellers who choose competing alternatives. This can be achieved in several ways—for example, by limiting interoperability between the gatekeeper’s primary product and rival ancillary services, restricting competitors’ ability to improve their quality, or steering buyers away from sellers that rely on competing offerings. Crucially, although ancillary services are purchased by sellers, they affect the quality and terms of interactions with buyers, thereby influencing buyer surplus too.

First, we show that the gatekeeper has an incentive to engage in self-preferencing even when its ancillary service is of lower quality than rival offerings—provided its quality is not too low for self-preferencing to remain profitable. This result contrasts with the Chicago School’s critique of conventional foreclosure, which argues that dominant firms typically lack incentives to exclude more efficient competitors. We find this result in a benchmark setting without network effects, and we also show that positive network effects may further strengthen the gatekeeper’s incentive to self-preference, helping explain why this strategy has become pervasive in today’s digital markets.

Self-preferencing allows the gatekeeper to charge a higher price for its ancillary service and gain a larger share of the adjacent market. However, given the primary product price, this also raises transaction costs for sellers, discouraging their participation in the primary market. Lower seller participation, in turn, reduces seller demand and diminishes the gatekeeper’s profits from the primary product. Thus, self-preferencing remains profitable only when the ancillary service’s quality is high enough for the gains in the adjacent market to outweigh the losses from reduced seller participation in the primary market.

Second, the welfare implications of self-preferencing are complicated. The practice can either benefit or harm buyers and sellers, depending on several factors, including the preferences of buyers and sellers for ancillary services and the gatekeeper’s pricing instruments. Notably, when the gatekeeper charges only sellers, its incentive to self-preference can align with sellers’ short-run interests. In such cases, sellers actually benefit when self-preferencing is profitable for the gatekeeper. However, the impact on buyers may be negative if they derive lower utility from the gatekeeper’s ancillary service

compared to rival offerings.

The intuition is as follows: after self-preferencing, each additional seller attracted to the market is more likely to choose the gatekeeper’s ancillary service, generating higher profits for the gatekeeper. This incentivizes the gatekeeper to lower the price of the primary product to stimulate seller participation, which in turn benefits sellers. As for buyers, if they value the gatekeeper’s ancillary service, more transactions will be fulfilled through the services they prefer following self-preferencing, making them better off. However, if they prefer rival ancillary services, they are worse off.

When the gatekeeper charges both sides of the market, buyers and sellers can either benefit from or be harmed by self-preferencing. In this context, the gatekeeper’s self-preferencing incentive is jointly determined by the preferences of both buyers and sellers for ancillary services. Hence, sellers can also be harmed by self-preferencing, even when it is profitable for the gatekeeper. This occurs when buyers strongly prefer the gatekeeper’s ancillary services, thereby strengthening the gatekeeper’s self-preferencing incentive, while sellers dislike those services.

These theoretical results provide valuable insights into the earlier evidence from the Google case in France. The FCA’s intervention focused solely on the publisher side, without considering the advertiser side. According to our model, such a remedy would reduce the surplus for both publishers and advertisers when advertisers dislike interacting with publishers using non-Google ancillary adtech. In this scenario, the FCA’s remedy pushed publishers toward non-Google technologies that advertisers were less willing to engage with, discouraging their participation. As a consequence, the reduction in the pool of advertisers diminished the surplus for publishers from these interactions, which in turn lowered the surplus for both parties. This understanding is supported by recent investigations by the DoJ and the EC, which adopt a broader approach by analyzing not only publishers but also advertisers in the display advertising market. Both investigations highlight that the majority of advertisers now rely on Google’s adtech for ad buying and that Google has made it increasingly difficult for advertisers to engage with publishers’ non-Google adtech—an observation that aligns with the insights from our model.

In the final part of the paper, we illustrate how the model can be applied for empirical evaluation. Using data from the adtech industry, we offer novel evidence on the ex post impacts of the FCA intervention and then re-assess policy counterfactual related to it. Our estimates indicate that French advertisers derive significantly more value from publishers using Google’s adtech than from those relying on rival technologies. Consistent with the model, we confirm a decline in total welfare following the FCA regulation. However, this welfare loss could have been mitigated by an alternative behavioral remedy simultaneously targeting Google’s restrictions on the advertiser side. Specifically, a modest

improvement in the relative value of third-party adtech for advertisers would have been sufficient to offset the welfare decline. We also consider a structural remedy that separates Google’s primary product and ancillary service, and compare it with the FCA’s behavioral intervention.³ Neither approach dominates *ex ante*: their effectiveness depends on how well they realign incentives and cross-side compatibility. Structural remedies can outperform behavioral ones when they significantly improve the competitiveness of rival adtech for both publishers and advertisers.

This study makes three key contributions. First, it develops a tractable yet flexible model that captures the fundamental mechanisms and trade-offs driving cross-market self-preferencing, and clarifies conditions under which it arises, its welfare effects, and its response to regulation. Second, it introduces new empirical evidence from a natural experiment in the French adtech market, shedding light on why well-intentioned remedies may backfire. Third, it leverages this framework to run counterfactual simulations, discussing the welfare implications of alternative behavioral and structural remedies and identifying conditions under which each becomes effective. Taken together, our findings yield a clear policy message: there is no universally superior remedy for self-preferencing. Effective regulation must recognize the two-sided nature of digital platforms and target both sides of the market simultaneously; otherwise, interventions risk fragmenting the market and reducing overall efficiency. By combining theory, empirical evidence, and counterfactual analysis, this paper offers the first integrated assessment of the welfare implications of self-preferencing in the adtech industry and informs the ongoing debate on how to regulate gatekeepers in digital markets.

The remainder of the paper proceeds as follows. Section 2 develops the model. Section 3 explores extensions including network effects and pricing instruments. Section 4 presents the empirical analysis and counterfactual simulations. Section 5 concludes.

Contribution to Literature

This paper intersects with four strands of literature. First, it contributes to the research on self-preferencing in digital markets (De Corniere and Taylor, 2014, 2019; Kang and Muir, 2022; Dendorfer, 2024; Motta, 2023), which largely focuses on hybrid-mode platforms.⁴ In contrast, we

³Behavioral interventions focus on regulating specific practices or conduct, without altering a firm’s core structure, while structural interventions involve changes to the market (e.g., creating the conditions for the entry of new players) or a firm’s organization (e.g., breaking up monopolies or forcing divestitures) to reduce anti-competitive power.

⁴Hybrid-mode platforms are defined as firms which serve as both intermediaries that facilitate interactions between buyers and third-party sellers, and simultaneously as retailers that sell directly to buyers. Self-preferencing on these platforms often involves recommendation bias toward the platform’s own offers. For instance, Zennyo (2022) models inclusion in consumers’ consideration sets; Kittaka and Sato (2022) assume platform offers are checked first; Hagi et al. (2022) consider suppressing innovative sellers.

study “cross-market” self-preferencing, where the platform favors its ancillary service by degrading the performance or value of rival services in adjacent markets. Moreover, although the choice and purchase of ancillary services are determined by one side of the market, these services are used to fulfill interactions with the other side. Therefore, the choice of ancillary service involves both sides of the market. Such “two-sided” market features highlight novel trade-offs and welfare implications: in the conventional research, hybrid-mode platforms competes directly with third-party sellers and can extract their surplus by charging referral fees. Hence, the platform faces a trade-off between monetizing through referral fees on third-party sales or earning retail margins from its own direct sales. However, in cross-market self-preferencing settings, the former monetization channel—specifically, the ability to extract a commission fee from third-party sales—is absent,⁵ and, contrary to the existing literature (Anderson and Bedre Defolie, 2024; Zenno, 2022; Etro, 2023; Padilla et al., 2022), welfare impacts of cross-market self-preferencing no longer hinge on referral fees.

Additionally, our paper suggests that a gatekeeper may engage in self-preferencing even when its ancillary service is inferior to that of its rivals, which contrasts with the Chicago School critique. Existing research (Fumagalli et al., 2018) indicates that the breakdown of Chicago School critique may arise from the gatekeeper’s imperfect ability to extract profit from third-party sellers (Hart et al., 1990; Rey and Tirole, 2007), concerns about future entry into the primary market by competing ancillary services (Fumagalli and Motta, 2020), or the gatekeeper facing competition in the primary product market (Ordover et al., 1990). Complementary to these channels, we highlight that self-preferencing incentives can also arise from the joint impact of buyers’ and sellers’ preferences, as well as the gatekeeper’s pricing instruments.

Second, our research relates to the literature on tying and bundling in digital markets. Traditional research on tying typically focuses on its effects in terms of price discrimination (Adams and Yellen, 1976; Bakos and Brynjolfsson, 1999), cost savings (Tirole, 2005; Hovenkamp and Hovenkamp, 2012), and the leveraging of market power (Stigler, 1963; Nalebuff, 2004; De Corniere and Taylor, 2021). Studies in digital markets suggest, however, that tying may also serve as a mechanism to encourage multihoming (Choi, 2010) or circumvent non-negative price constraints (Amelio and Jullien, 2012; Choi and Jeon, 2021; Bisceglia and Tirole, 2023). Our approach diverges from past research in two key aspects. First, we analyze a setting where the ancillary service is not tied to the primary product. Rather than altering the choice set, the focus is on quality changes—sellers using rival ancillary services experience relatively lower performance due to self-preferencing. This is consistent with empirical observations on platforms like Google, where publishers using non-Google adtech can still interact

⁵For example, Amazon cannot charge warehouses storing goods for third-party sellers; Apple cannot extract fees from competing subscription services or payment technologies; Google cannot charge competing adtech providers for interacting with Google’s tools in other layers of the adtech stack.

with Google’s adtech in other layers, but their performance may be constrained by Google.

Also, while most existing studies focus on tying at the consumer level, our focus shifts to the seller side.⁶ In our model, sellers choose the ancillary service, while buyers passively accept and are affected by the seller’s decision. This additional layer introduces a crucial two-sided market feature and adds novelty to literature with important regulatory implications.

Third, the paper contributes to the broader literature on multi-sided markets (Belleflamme and Peitz, 2021; Teh, 2022; Bisceglia and Tirole, 2023; Crawford et al., 2023; Rochet and Tirole, 2004; Jullien, 2005; Biglaiser et al., 2019; Crawford et al., 2023; Bergemann and Bonatti, 2024). Network effects play a critical role: our model shows that strong cross-group network effect may raise the platform’s incentive to self-preference and reduce the risk of welfare loss.

Lastly, the paper is relevant to studies on platform recommendation and search (Hagiu and Jullien, 2011; Inderst and Ottaviani, 2012; Tadelis, 2016; Hunold and Muthers, 2017; Shen and Wright, 2019; Li et al., 2020; Long and Liu, 2023; Ostrovsky, 2023; Gambato and Risco, 2024), and to empirical work on self-preferencing, mostly focusing on Amazon (Lam, 2021; Gutiérrez, 2022; Farronato et al., 2023; Waldfogel, 2024).⁷

2 Benchmark Model

This section develops a benchmark model to capture the economic dynamics of cross-market self-preferencing. The framework is deliberately simplified to focus on the core market feature: the two-sided nature of preferences for ancillary services from both buyers and sellers, that is, the two-sidedness of ancillary service choice in affecting surplus for both parties. Building on this benchmark, we next incorporate another two-sided market feature from the conventional research: network effects, where the surplus of users in one group is positively affected by the participation of users in the other group. Moreover, we introduce additional factors in Section 3 and the Appendix, including buyer charging, varied consideration sets, and other extensions. We then analyze how the conclusions evolve as each factor is introduced, allowing us to isolate the impact of each.

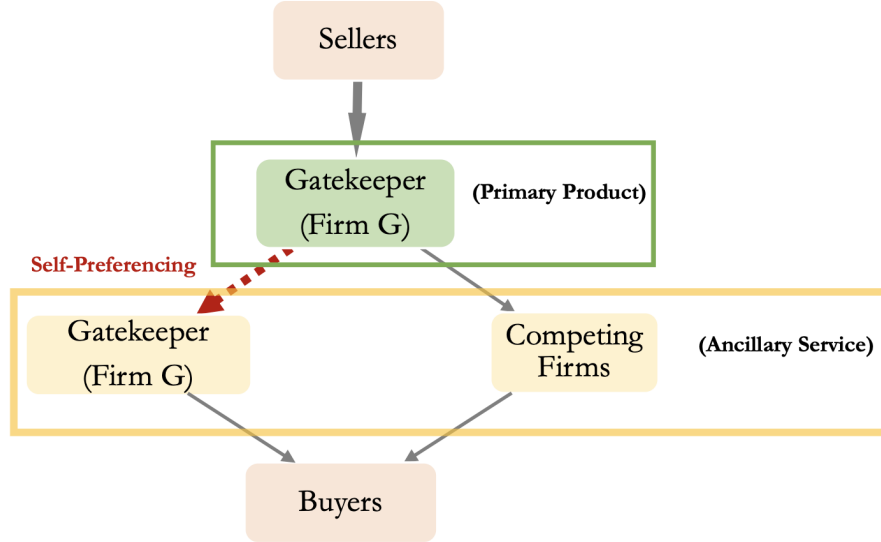
⁶One exception is De Corniere et al. (2024), which examines the tying of ancillary services for sellers in the context of Amazon and point out a mechanism where tying serves to diminish product differentiation.

⁷Chen and Tsai (2024) find Amazon products receive more cross-sell exposure; Lee and Musolf (2021) show that Amazon recommendations are price elastic and may increase consumer surplus.

Model Setup

There is a two-sided market where users gain value from interacting with each other. To simplify notations, we refer to one group of users as sellers (he/his) and the other as buyers (she/her). In the context of the display advertising market, the two groups correspond to publishers (sellers of advertising slots) and advertisers (buyers of advertising slots).

Figure 2: Market Structure



Sellers rely on a primary product to interact with buyers. In addition, they must also purchase an ancillary service in an adjacent market to facilitate interactions. As shown in Figure 2, a gatekeeper, denoted as firm G , sells both in the primary product and ancillary service markets. While the company monopolizes the primary (product) market, it faces competition in the ancillary service market. In addition to firm G , there are also $N \geq 1$ competitive fringe firms selling similar ancillary services. Although our model is largely motivated by the Google case, the overall market structure generally applies to several recent antitrust cases as illustrated in Table 1. Detailed descriptions of these cases are provided in the Appendix.

Table 1: Mapping between Model Setup and Recent Antitrust Cases

Gatekeeper	Buyers	Sellers	Primary Product	Ancillary Service	Self-Preferencing Practice
Google	Advertiser	Publisher	Google DoubleClick	Supply-Side Platform	Interoperability or Data Restrictions
Amazon	Consumers	Retailers	Amazon Marketplace	Storage and Shipping	User Steering
Apple	Mobile Users	App Developers	iOS App Store	Electronic Payment Technology	Marketing Restrictions

Within this market structure, the gatekeeper can practice self-preferencing across markets, leveraging its dominant position in the primary product market to favor its sales in the ancillary service market. To achieve the goal, the gatekeeper firm diminishes the relative value of competing ancillary services, while making its own ancillary service more favorable. In digital markets, gatekeeper firms can employ varied approaches to practice such self-preferencing. For instance, Google can restrict data access for rival adtechs, making it difficult for them to understand why they lost in advertising auctions and hindering their ability to improve. This, in turn, reduced the performance and quality of competing services. Also, Google can reduce the interoperability between its technology and that of others, so that once users adopt Google's dominant adtech in one layer, the value of using non-Google adtech in another layer diminishes.

Fringe firms all incur the same entry cost L , and their services have no vertical differentiation.⁸ Specifically, we refer to the gatekeeper firm G 's ancillary service in the adjacent market as the first-party (1P) service and all other firms' ancillary services as third-party (3P) services. Accordingly, we categorize sellers who use the 1P service as 1P sellers and sellers who use 3P services as 3P sellers. To simplify without loss of generality, we normalize the marginal costs of the primary product and ancillary services to zero throughout the analysis.

A seller has two ancillary services in his consideration set: one is always the 1P service from firm G , and the other is a randomly chosen 3P service. Hence, the probability that any fringe firm $k \in \{1, 2, \dots, N\}$ is included in the seller's consideration set equals $1/N$. In the Appendix, we relax this assumption and analyze the scenario where all services in the consideration set are chosen randomly. In this case, fringe firms not only compete with the gatekeeper firm but also with each other. We show that the conclusions drawn in this market are qualitatively similar to those in the benchmark.⁹

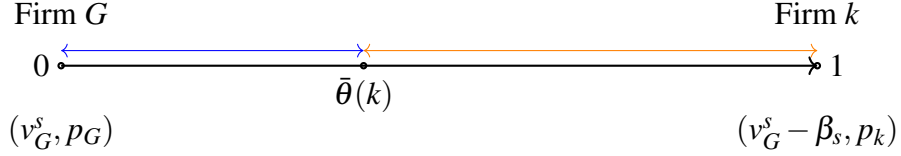
A seller perceives the two ancillary services in his consideration set as both vertically and horizontally differentiated. Mapping to the adtech market, vertical differentiation arises from the varying quality of algorithms among adtech companies, while horizontal differentiation stems from factors such as specialization in ad formats or targeting capabilities. For instance, some adtech companies may specialize in image ads, while others focus on video ads; some may emphasize behavioral targeting, while others specialize in demographic or geographic targeting. Due to horizontal differentiation, market share is not dominated by a single player, as seen in the adtech market, where Google's ancillary adtech captures approximately 50% of ad sales in the US.

⁸This entry cost is not collected by firm G .

⁹Our key findings regarding welfare also remain valid in the nested demand scenario, where sellers first choose between 1P and 3P ancillary services, and if they choose 3P ancillary services, they then determine which fringe seller to purchase from.

Vertical differentiation is modeled by sellers' diverse profits/values when they use different ancillary services alongside the primary product to fulfill interactions. If sellers choose the 1P service, their profit from joining the market is denoted as v_G^s . However, if sellers choose a 3P service from a competing firm, their profit is denoted as $v_k^s = v_G^s - \beta_s$. The parameter β_s measures the vertical difference between the two ancillary services for sellers. Its value can be either positive, zero, or negative. If 1P sellers derive a higher profit from joining the market than 3P sellers, then $\beta_s > 0$. Otherwise, it is the opposite.

Figure 3: Hotelling Line



Horizontal differentiation is modeled in the Hotelling fashion. As shown in Figure 3, the two ancillary services are located at the two ends of a unit-length Hotelling line, with the 1P service located at 0 and the 3P service located at 1. Sellers are uniformly distributed on this Hotelling line and incur the cost of not having one's ideal ancillary service that linearly increases in distance at rates t . Sellers can fulfill transactions through only one ancillary service, namely, multihoming is not allowed. In the advertising market, publishers may affiliate with multiple adtech companies simultaneously, but each transaction is ultimately fulfilled through only one adtech. To ensure tractability, we abstract away the bidding competition among adtech companies and view the adtech that ultimately fulfills the publisher's interaction with the advertiser as the publisher's choice. Within this setup, the surplus of seller j , who has fringe firm k in his consideration set and purchases the ancillary service from firm $i \in \{G, k\}$, is given by

$$U_s(\theta_j, i) = v_i^s - p_i - m_S - t|\theta_j - l_i|,$$

where $\theta_j \in [0, 1]$ is the position of seller j , $l_i \in \{0, 1\}$ represents the location of firm i , m_S is the primary product price, and p_i denotes the corresponding ancillary service price.

Although sellers are the ones who choose the ancillary services to be used with their primary product, their choices are later used to facilitate interactions with buyers. Thus, the selection of ancillary services affects not only the profit of sellers but also the surplus of buyers. To capture the two-sided nature in preferences, buyers' utility is denoted as v_G^b if sellers use the 1P service to fulfill interactions, and as $v_C^b = v_G^b - \beta_b$ if sellers employ 3P services. The parameter β_b measures the vertical difference between the two ancillary services for buyers. This value differentiation can arise from fair competition between firms, which leads buyers to intrinsically prefer interacting with one firm over another, or it

could result from companies engaging in unfair competition or foreclosure practices that strategically alter buyers' perceptions over products. If $\beta_b = 0$, buyers are completely indifferent among all sellers. If $\beta_b > 0$ ($\beta_b < 0$), buyers expect a higher (lower) utility from 1P sellers than from 3P sellers. There is no correlation between β_b and β_s , meaning that the preferences of buyers and sellers for ancillary services can either differ or be identical. Such flexibility allows our model to be applied to different market contexts. To simplify, we assume that there is no horizontal differentiation between ancillary services for buyers. Such assumption does not qualitatively affect our key conclusions in the benchmark.¹⁰

In summary, a buyer's utility based is given by:

$$U_b = \begin{cases} v_G^b, & \text{if 1P service;} \\ v_G^b - \beta_b, & \text{if 3P service.} \end{cases}$$

The timeline of the game is as follows: First, firm G sets its price in the primary market. Second, fringe firms and firm G simultaneously choose the prices of their own ancillary services. Given the prices in the primary and adjacent markets, buyers and sellers make entry decisions. Last, sellers who join the market purchase the ancillary service that gives them the highest value. Based on sellers' choice of ancillary services, buyers and sellers realize corresponding surpluses in the market. To prevent trivial market equilibria in the analysis, we make the following assumptions, with the explicit equations for all cutoff values provided in the Appendix.

Assumption 1. *In the benchmark model, we assume that:*

- I) [Positive Demand] the ancillary service values satisfy $|\beta_s| < 3t$ so that both the 1P service and 3P service derive a positive demand at equilibrium;*
- II) [Competitive Market] the surplus is sufficiently large $v_G^s > \tilde{v}_G^s$, such that there exists competition between the 1P service and 3P services; and*
- III) [Market Participation] the surplus satisfies $\underline{v}_G^b < v_G^b < \bar{v}_G^b$ and $\underline{v}_G^s < v_G^s < \bar{v}_G^s$, so that user entry does not hit corner solution.*

Our benchmark setup focuses on a simple setting to highlight underlying mechanisms. Several extensions are provided in the later sections. First, the benchmark abstracts away the cross-group

¹⁰Specifically, introducing horizontal differentiation for buyers in the same manner as for sellers does not qualitatively change our key findings, but it can quantitatively increase the β_b threshold beyond which buyers benefit from self-preferencing, shifting it from 0 to $\frac{\beta_s}{3}$.

network effects. In other words, a user's surplus—both on the buyer side and the seller side—is not influenced by the participation of the other group. Such network effects are incorporated into the model in Section 3, where we investigate the influence of network effects on self-preferencing incentives and its welfare implications. Second, in the benchmark, we assume that buyers enjoy free services to facilitate their interactions. This assumption is relaxed in Section 3, where we examine the market outcomes when the gatekeeper charges both buyers and sellers.

Other extensions and robustness checks are detailed in the Appendix. Specifically, we examine the case of horizontal differentiation between 1P and 3P services to be influenced by firms' entry into the adjacent market (Appendix B.4), explore a fully random consideration set (Appendix B.5), relax the uniform distribution of outside options (Appendix B.6), investigate alternative definitions of self-preferencing (Appendix B.7), and relax the assumption that the gatekeeper firm takes market size as given when determining the 1P ancillary service price (Appendix B.8).

Market Equilibrium

We next proceed to solve for the subgame perfect equilibrium. We initiate the analysis by examining sellers' demands for ancillary services, as well as the demands for the primary product on both sides of the market. We then investigate firms' pricing decisions in the adjacent market, followed by an exploration of firm G 's profit-maximizing pricing strategy in the primary market.

Seller Demands If firm G and firm $k \in \{1, 2, \dots, N\}$ are in a seller's consideration set, the seller is indifferent between the two firms when his location $\bar{\theta}(k)$ satisfies $U_S(\bar{\theta}(k), G) = U_S(\bar{\theta}(k), k)$. Solving this equation, we derive the indifferent seller's location as follows:

$$\bar{\theta}(k) = \frac{\beta_s + p_k - p_G + t}{2t}.$$

As illustrated in Figure 3, sellers located at $\theta_j < \bar{\theta}(k)$ will purchase from firm G , while the rest will choose firm k . Given that fringe firms have an equal probability of being included in a seller's consideration set, the fraction of sellers considering firm k is n_s/N . As for firm G , it is considered by all sellers, regardless of their consideration sets. Therefore, the demands for the two ancillary services

can be expressed as:

$$Q_G = \sum_{k=1}^N \bar{\theta}(k) \frac{n_S}{N} = \sum_{k=1}^N \frac{(\beta_s + p_k - p_G + t)}{2t} \frac{n_S}{N},$$

$$Q_k = (1 - \bar{\theta}(k)) \frac{n_S}{N} = \left(1 - \frac{\beta_s + p_k - p_G + t}{2t}\right) \frac{n_S}{N},$$

where Q_G represents the demand for the 1P service, and Q_k denotes the demand for firm k .

User Entry Motivated by the intricate rules governing digital markets, we assume that sellers become aware of their consideration set and their specific location on the Hotelling line *after* joining the market. This corresponds to the idea that the rules of the digital advertising market are complex, and it takes time for publishers to fully understand the interactions between multiple layers of adtech. Only after a seller joins the market and learns how the ancillary service interacts with the primary product does he become aware of his preferences. In such a context, a seller's expected surplus from joining the market, denoted as $E[U_s]$, equals:

$$E[U_s] = \sum_{k=1}^N \left(\int_0^{\bar{\theta}(k)} (v_G^s - p_G - tx) dx + \int_{\bar{\theta}(k)}^1 (v_G^s - \beta_s - p_k - t(1-x)) dx \right) \frac{1}{N} - m_s. \quad (1)$$

As for buyers, they interact with the ancillary services chosen by sellers. This reflects the observations that advertisers in digital advertising markets must interact with publishers' adtech to fulfill transactions over advertising slots. Consequently, their expected utility depends on the distribution of sellers across ancillary services and can be expressed as:

$$E[U_b] = \sum_{k=1}^N \left(\int_0^{\bar{\theta}(k)} v_G^b dx + \int_{\bar{\theta}(k)}^1 (v_G^b - \beta_b) dx \right) \frac{1}{N}.$$

Users in both groups will join the market if they expect to derive a higher surplus than their outside options. To ensure tractability, we assume that both buyers' and sellers' outside options follow a uniform distribution in the interval $[0, 1]$, and their user sizes are normalized to one.¹¹ Within this model setup, the primary product market participation in both groups are given by: $n_S = E[U_s]$ and $n_B = E[U_b]$.

¹¹The relaxation of uniform distribution is discussed in the Appendix.

Firm Pricing Given the demand functions, we now investigate firms' pricing strategies in the ancillary service market. For fringe firm k , its profit comes solely from selling ancillary services and can be calculated as:

$$\pi_k = p_k Q_k - L = p_k \frac{(-\beta_s + p_G - p_k + t)n_S}{2Nt} - L.$$

As for firm G , it earns additional profit from providing the primary product. Accordingly, its profit is expressed as:

$$\pi_G = m_S n_S + p_G \sum_{k=1}^N \bar{\theta}(k) \frac{n_S}{N} = m_S n_S + \sum_{k=1}^N p_G \frac{(\beta_s - p_G + p_k + t)n_S}{2Nt}.$$

As illustrated in the previous analysis, sellers' demand for market participation is jointly determined by all firms in the ancillary service market. To simplify, we assume each firm in the ancillary service market has minimal impact and takes the aggregate demand n_S as given.¹² At equilibrium, the ancillary service prices are given by:

$$p_G = t + \frac{\beta_s}{3}, \quad (2)$$

$$p_C = t - \frac{\beta_s}{3}, \quad (3)$$

where p_C represents the price set by all fringe firms at the symmetric equilibrium.

Plugging in the equilibrium prices, the ancillary service demands are equal to:

$$Q_G = \left(\frac{1}{2} + \frac{\beta_s}{6t}\right)n_S, \quad (4)$$

$$Q_C = \left(\frac{1}{2} - \frac{\beta_s}{6t}\right)\frac{n_S}{N}, \quad (5)$$

where Q_C denotes the demand for each fringe firm at the symmetric equilibrium. Accordingly, the equilibrium ratio of sellers choosing the 1P service is denoted as $q_G = Q_G/n_S = \frac{1}{2} + \frac{\beta_s}{6t}$.

Primary Product Pricing Lastly, we analyze firm G 's pricing strategy over its primary product. Given the ancillary service prices (expression 2-3) and demands (expression 4-5) at equilibrium, the explicit

¹²In the Appendix, we relax this assumption and derive qualitatively similar key findings.

demands for market participation can be expressed as:

$$n_B = v_G^b + \frac{\beta_b(\beta_s - 3t)}{6t}, \quad (6)$$

$$n_S = \frac{\beta_s^2 - 18\beta_s t - 36tm_S - 45t^2 + 36tv_G^s}{36t}. \quad (7)$$

Maximizing firm G 's profit by solving the first-order condition, we derive the profit-maximizing price in the primary market as:

$$m_S^* = \frac{36tv_G^s - \beta_s^2 - 30\beta_s t - 63t^2}{72t}. \quad (8)$$

Given the equilibrium pricing strategies, the demands for the primary product are given by:

$$n_B^* = \frac{\beta_b\beta_s + 6tv_G^b - 3\beta_b t}{6t}, \quad (9)$$

$$n_S^* = \frac{\beta_s^2 - 2\beta_s t - 9t^2 + 12tv_G^s}{24t}, \quad (10)$$

and the resulting demands for ancillary services are $Q_G^* = (\frac{1}{2} + \frac{\beta_s}{6t})n_S^*$ and $Q_C^* = (\frac{1}{2} - \frac{\beta_s}{6t})\frac{n_S^*}{N}$.

We lastly calculate the equilibrium profits of firms and derive:

$$\pi_G^* = \frac{(\beta_s^2 - 2\beta_s t - 9t^2 + 12tv_G^s)^2}{576t^2}, \quad (11)$$

$$\pi_C^* = \frac{(3t - \beta_s)^2(\beta_s^2 - 2\beta_s t - 9t^2 + 12tv_G^s)}{432Nt^2} - L, \quad (12)$$

where π_C^* denotes the profit of each individual fringe firm at the symmetric equilibrium.

In summary, the market equilibrium in the benchmark can be summarized as in Proposition 1:

Proposition 1. *Under Assumption 1, there is a unique equilibrium such that a total of n_B^* buyers and n_S^* sellers, as shown in equations (9) to (10), join the market. Among them, a portion of $(3t + \beta_s)/6t$ opts for 1P services, while a proportion of $(3t - \beta_s)/6t$ chooses 3P services at the subgame-perfect equilibrium. The fringe firms and firm G choose the pricing strategies listed in expression (2)-(3) and (8), leading to the profits as shown in equations (11)-(12).*

Self-Preferencing Impact

Based on the market equilibrium, we now examine the impact of self-preferencing through comparative statics, with all proofs provided in the Appendix. In many markets, to avoid detection by competition authorities, gatekeepers typically implement marginal rather than substantial adjustments. We therefore focus on how the market evolves when there is a slight reduction in sellers' relative value of 3P services compared to 1P services. To isolate this reallocation effect, we assume that the total value of sellers using the two ancillary services remains unchanged following self-preferencing. Mapping this adjustment into our model, self-preferencing is formally defined as follows:

Definition 1. *Self-preferencing refers to a practice by which the gatekeeper increases the relative value of its 1P ancillary service compared to competing 3P ancillary services, while holding constant the total value $v^s = v_G^s + v_k^s$. In our framework, self-preferencing corresponds to a marginal increase in the vertical differentiation parameter β_s , such that $v_G^s = (v^s + \beta_s)/2$ and $v_k^s = (v^s - \beta_s)/2$*

As defined, self-preferencing shifts value from the 3P ancillary service to the 1P ancillary service. In other words, if self-preferencing reduces sellers' value of using 3P services, it results in a corresponding increase in their value of using 1P services. To illustrate, consider an example where a website sells advertising slots through two complementary layers of adtech, one of which is dominated by Google. If Google engages in self-preferencing by restricting rivals' data access in the ancillary services and limiting their algorithm performance, more transactions — and thus more value — shift from 3P ancillary adtech to Google's 1P ancillary adtech, inducing a value transfer.

We recognize that, in some contexts, self-preferencing not only induces a reallocation effect but also impacts the total transaction value v^s . Continuing with the Google adtech example, this would correspond to a scenario where self-preferencing leads to a decrease in the total value of using adtech. Motivated by this possibility, we discuss alternative definitions of self-preferencing in the Appendix. We show that when self-preferencing affects both total value of adtech and its distribution, the key conclusions regarding welfare in the benchmark remain valid, but the gatekeeper's motivation for self-preferencing may be affected.

Building on Definition 1, we then investigate the implications of self-preferencing by conducting comparative statics with respect to β_s . As shown in Equations (2-3), the 1P and 3P ancillary services have the same price when there is no vertical differentiation for sellers. The introduction of vertical differentiation, β_s , lowers the price of 3P services by $\beta_s/3$ while raising the 1P service price by $\beta_s/3$. If the gatekeeper firm practices self-preferencing, the 1P service becomes relatively more attractive, i.e., β_s increases. Hence, in the adjacent market, the 1P service price increases, while the price of 3P services decreases.

In the primary market, firm G diminishes its product price following self-preferencing. This is because the 1P service becomes more expensive after self-preferencing, making it more costly for sellers to fulfill transactions in the market. As a consequence, sellers' entry is discouraged, resulting in a reduction in their demand for the primary product. To compensate for this negative impact, the gatekeeper firm G has to reduce its product price m_S in the primary market.

In summary, the impact of self-preferencing on firms' pricing strategies in the primary and adjacent markets can be outlined as follows:

Corollary 1. *In the benchmark, practice of self-preferencing:*

- I) increases the 1P service price p_G ;*
- II) decreases the 3P service price p_C ; and*
- III) decreases the primary product price m_S .*

We then examine the gatekeeper's profit to shed light on the incentives for self-preferencing. We find that competing firms always suffer from self-preferencing. However, for the gatekeeper firm, engaging in self-preferencing may be either profitable or not, as summarized in Corollary 2.

Corollary 2. *In the benchmark, the practice of self-preferencing always reduces the profit of competing firms. However, self-preferencing can either increase or decrease firm G 's profits. Specifically, self-preferencing decreases firm G 's profit when $\beta_s < -2t$ and increases it otherwise.*

The underlying mechanisms are as follows. There are several conflicting impact following self-preferencing. In the adjacent market, self-preferencing makes the 1P service more attractive. As a result, competing firms experience a lower market share and are forced to reduce their service prices. Both effects diminish the competing firms' profits. However, for the gatekeeper firm, the opposite occurs. Self-preferencing allows the gatekeeper to charge a higher price, capture a larger market share, and thus achieve a greater profit in the ancillary service market.

In the primary market, a larger proportion of sellers are forced to use the 1P service at a higher price following self-preferencing. With the same primary product charge, this reduces the profitability of sellers entering the market and discourages their participation. The resulting demand reduction for the primary product, in turn, lowers the gatekeeper firm's profits in the primary market. When the value of 3P services is significantly higher than that of the 1P service, most sellers are 3P sellers who incur a surplus loss after self-preferencing. As a result, the gatekeeper firm must substantially reduce the

primary product price to offset sellers' loss. In this context, the negative impact becomes dominant, making self-preferencing unprofitable for the gatekeeper.

Interestingly, our results also suggest that the gatekeeper firm may be motivated to engage in self-preferencing even when its ancillary service is inferior to that of rivals (i.e., $\beta_s < 0$). This contrasts with the conventional Chicago School critique, which argues that dominant firms rarely have the incentive to foreclose more efficient rivals. Moreover, as discussed in the subsequent analysis, this incentive can be further influenced by the gatekeeper's pricing strategies and the preferences for ancillary services on both sides of the market.

Turning to the demand for market participation, we find that self-preferencing can lead to either higher or lower participation, depending on both buyers and sellers' preference over ancillary services. Precisely, the results can be summarized in the following corollary:

Corollary 3. *In the benchmark, self-preferencing can either increase or decrease the demand for market participation. Specifically,*

- I) if 1P sellers derive much lower surplus than 3P sellers, i.e., $\beta_s < -2t$, then self-preferencing decreases sellers' demand for market participation; Otherwise, it is the opposite;*
- II) If buyers prefer 1P sellers, i.e., $\beta_b > 0$, then self-preferencing increases buyers' demand for market participation; Otherwise, it is the opposite.*

When a gatekeeper firm engages in self-preferencing, several conflicting effects emerge regarding seller participation. The underlying mechanisms can be illustrated by revisiting the sellers' entry decisions. Specifically, a seller's surplus from entering the market can be simplified as follows:

$$\begin{aligned}
 E[U_s] &= \sum_{k=1}^N \left(\int_0^{\bar{\theta}} (v_G^s - p_G - tx) dx + \int_{\bar{\theta}}^1 (v_G^s - \beta_s - p_k - t(1-x)) dx \right) \frac{1}{N} - m_s, \\
 &= -m_s + \underbrace{\int_0^{q_G} \left(\frac{v^s}{2} - t + \frac{\beta_s}{6} - tx \right) dx}_{\text{sellers using 1P service}} + \underbrace{\int_{q_G}^1 \left(\frac{v^s}{2} - t - \frac{\beta_s}{6} - t(1-x) \right) dx}_{\text{sellers using 3P service}}, \tag{13}
 \end{aligned}$$

after plugging in the ancillary prices (Equations 2 and 3) and v_G^s as defined previously (Definition 1).

As illustrated in Equation 13, self-preferencing reduces the surplus of 3P sellers while increasing the surplus of 1P sellers. Additionally, as demonstrated in Corollary 1, self-preferencing motivates the gatekeeper to lower the primary product price, thereby enhancing the surplus of sellers. When 3P

services become more attractive, a larger proportion of sellers choose 3P services. This increased share of 3P sellers amplifies the negative effects of self-preferencing, ultimately leading to an overall reduction in seller participation when β_s falls below a certain threshold.

As for buyers, their market participation can be either encouraged or discouraged after self-preferencing. As illustrated in the previous analysis, self-preferencing encourages more sellers to choose the 1P service. When buyers prefer the 1P service over 3P services for fulfilling interactions, self-preferencing drives more sellers to switch to the ancillary service that buyers favor. In this case, buyers are better off and more inclined to enter the market. Conversely, when buyers strongly prefer 3P sellers over 1P sellers, self-preferencing steers more sellers toward the ancillary services that buyers dislike. As a result, buyers are discouraged from entering the market following self-preferencing.

Note that within the model setup, the total surplus of sellers and buyers can be expressed as:

$$\begin{aligned} Surplus_S^* &= \int_0^{n_S} E[U_s]dO + \int_{n_S}^1 OdO = \int_0^{n_S} n_S dO + \int_{n_S}^1 OdO = \frac{1+n_S^2}{2}, \\ Surplus_B^* &= \frac{1+n_B^2}{2}, \end{aligned}$$

where O is the outside option. Since the user surplus in both groups is increasing in their participation, we could view market participation as a measure of user surplus in each group. Keeping this in mind, our results suggest that self-preferencing doesn't always diminish user surplus at equilibrium. Instead, it has the potential to benefit both buyers and sellers if specific conditions are satisfied. The finding is in surprising contrast to a common belief that self-preferencing is detrimental to the society. In the benchmark, we demonstrate that both buyers and sellers may achieve a higher surplus after self-preferencing.

The more interesting results emerge when combining Corollary 2 with Corollary 3. We find that the gatekeeper firm's incentive for self-preferencing is always aligned with the sellers' interests. As summarized in Corollary 4, sellers always benefit from self-preferencing when such practice is profitable for the gatekeeper. This occurs because, after self-preferencing, a larger proportion of sellers choose the 1P service. In other words, for each additional seller attracted to the market, the gatekeeper firm, on average, earns higher profits following self-preferencing. This enhances the dominant firm's incentive to attract more sellers, making it willing to lower the primary product price in exchange for a larger market size. Consequently, sellers benefit from this mechanism.

Corollary 4. *In the benchmark, the gatekeeper firm's incentive for self-preferencing is aligned with the sellers' surplus: when the gatekeeper firm finds self-preferencing profitable, the surplus of sellers always increases following self-preferencing, and vice versa. However, the impact of self-preferencing*

on buyers' surplus can be either positive or negative. Specifically:

- I) if $\beta_b > 0$, self-preferencing increases buyers' participation and their surplus;*
- II) if $\beta_b < 0$, self-preferencing reduces buyers' participation and their surplus.*

As for buyers, harmful impacts may arise even when the gatekeeper firm is motivated to practice self-preferencing. The findings highlight the potential negative effects of self-preferencing on welfare and clarify the conditions under which it may occur. More importantly, they emphasize the critical role of buyers' preferences in determining the impact of self-preferencing on social welfare.

The benchmark model developed in this section highlights the complex tradeoffs faced by regulators in addressing cross-market self-preferencing. In the following section, we first consider some extensions (with additional ones presented in the Appendix) and then examine a case of how such tradeoffs play out in practice, using the FCA's intervention in the Google adtech case as a case study.

3 Extensions

In this section, we explore three extensions. First, we examine the role of network effects in determining self-preferencing impact. Next, we study how the gatekeeper firm's access to different pricing instruments affects self-preferencing welfare implications. Third, we discuss the necessity of regulating self-preferencing in the long run.

Network Effects

As highlighted in the existing literature, a key characteristic of two-sided markets is the presence of network effects: a user's surplus from joining the market typically increases in the participation of the other group. In the context of the digital advertising market, this reflects the fact that publishers are likely to obtain higher bids/value when more advertisers join the market, and advertisers are likely to have better advertising performance when more publishers open advertising slots. Incorporating these positive cross-group network effects, we reach qualitatively similar welfare conclusions to those in the benchmark. When self-preferencing is profitable for the gatekeeper, sellers benefit, while buyers may be either better or worse off. However, after the introduction of network effects, there are changes in

the conditions under which the gatekeeper firm finds it profitable to engage in self-preferencing, as well as the circumstances where the harmful impact on buyers may emerge.

Following past literature, we capture positive network effects by the linear expression $e_s n_B$, where $e_s > 0$ measures the strength of network effects. Combining the network effects with sellers' surplus in the benchmark, the total surplus of seller j if using ancillary service from firm $i \in \{G, 1, 2, \dots, N\}$ is now equal to:

$$U_s(\theta_j, i) = v_i^s - p_i - m_S - t|\theta_j - l_i| + e_s n_B.$$

Similarly, on the buyer side, network effects are represented by $e_b n_S$, where $e_b > 0$ measures the magnitude of network effects derived from each seller. To guarantee profit concavity, we assume network effects satisfy $0 < e_b e_s < 1$. Accordingly, a buyer's total expected utility from joining the market is given by $U_b = v_i^b + e_b n_S$ if her transaction is fulfilled by the ancillary service from firm $i \in \{G, 1, 2, \dots, N\}$.

Following the steps as shown in the Appendix, we find that network effects can either increase or decrease the gatekeeper's incentive for self-preferencing, depending on buyers' preference over ancillary services.

Corollary 5. *In a two-sided market with positive network effects, network effects enhance the gatekeeper's incentive for self-preferencing when buyers prefer the 1P service over the 3P services, and reduce its incentive otherwise. Specifically, the practice of self-preferencing is profitable for firm G when $\beta_s > -2t - e_s \beta_b$, and non-profitable otherwise.*

Network effects make the participation of buyers and sellers positively correlated. As a consequence, the gatekeeper firm G 's strategy now is jointly determined by both sides of the market. If buyers exhibit a growing preference for 1P sellers over 3P sellers, they benefit from self-preferencing as it encourages more interactions to be fulfilled through their favored ancillary service. This positive impact on buyers, in turn, spreads to the seller side through network effects, generating an additional incentive for firm G to practice self-preferencing. Conversely, if buyers prefer 3P sellers over 1P sellers, self-preferencing leads more sellers toward the ancillary services that buyers dislike, thereby discouraging buyer participation. This negative impact on buyers diminishes seller entry, discouraging the gatekeeper firm's incentive for self-preferencing.

In summary, network effects can either strengthen or diminish the gatekeeper firm's incentive for self-preferencing, depending on the vertical differentiation between ancillary services for buyers. This result is intriguing because, on the one hand, it suggests that improving buyers' preference for the gatekeeper's ancillary service—either through quality enhancements or marketing efforts—will

increase the gatekeeper's incentive for self-preferencing. On the other hand, this suggests that the expectation of extracting future profits through self-preferencing may give the established and dominant company a strong incentive to offer and invest in high-quality ancillary services. In contrast, newer or smaller companies in the primary product market, or competing firms in adjacent markets, may lack such incentives. Hence, allowing self-preferencing in the short term does not necessarily harm consumers, as it can potentially enhance the gatekeeper's investment motivation in ancillary service quality and address the disconnect between market expansion in the primary market and the investment incentive in ancillary services.¹³

In addition to the gatekeeper's self-preferencing incentive, we also analyze welfare implications and derive Corollary 6.

Corollary 6. *In a two-sided market with positive network effects,*

- I) self-preferencing increases seller participation when $\beta_s > -2t - e_s\beta_b$ and decreases otherwise;*
- II) self-preferencing increases buyer participation when $\beta_b > -\frac{e_b(2t+\beta_s)}{2-e_be_s}$ and decreases otherwise.*

Combining the results in Corollary 5 and Corollary 6, we find qualitatively similar conclusions to the benchmark: the gatekeeper's motivation for self-preferencing aligns with the sellers' interests. However, the gatekeeper's motivation for self-preferencing can conflict with buyers' interests, even when buyers benefit from positive network effects.

Corollary 7. *In a two-sided market with positive network effects, the gatekeeper firm's incentive for self-preferencing is aligned with the sellers' surplus: when the gatekeeper firm finds self-preferencing profitable, the surplus of sellers always increases following self-preferencing, and vice versa. However, given that self-preferencing is profitable for the gatekeeper, its impact on buyers' surplus can be either positive or negative. Specifically, self-preferencing increases buyers' participation and their surplus when $\beta_b > -\frac{e_b(2t+\beta_s)}{2-e_be_s}$ and decreases otherwise.*

¹³This finding aligns with recent observations on Temu, a relatively new Boston-based, Chinese-owned shopping platform. Since its launch in 2022, Temu has invested heavily in expanding its market share, frequently holding the top spot for mobile downloads among both Apple and Android users over the past two years.¹⁴ Temu has faced significant criticism regarding the shipping of products sold on its platform, the "ancillary services". As pointed out by Time: "Temu has had trouble delivering within the promised time window."¹⁵ In 2023, the company had a C rating from the Better Business Bureau (BBB) and an average customer rating of 1.4 out of 5 stars. Unlike Amazon, Temu is relatively new and does not yet hold a dominant market share. As such, the company has little incentive to invest in its own ancillary services—namely, logistics networks—and instead relies heavily on external shipping providers like UPS, USPS, and FedEx. Temu's rapid market expansion has significantly increased the demand for shipping, with carriers reportedly handling around 900,000 packages per day in the U.S.¹⁶ However, external logistics providers lack sufficient investment incentive to keep up with this growth. As a result, consumers suffer from the disconnect between Temu's aggressive market expansion and the insufficient investment by external shipping companies to handle the increased demand.

Interestingly, Corollary 5 and 6 also suggest that the gatekeeper's incentive for self-preferencing may better align with social welfare, potentially generating superior market outcomes compared to a system in which sellers determine which ancillary service should be favored. Indeed, when network effects are strong, the gatekeeper internalizes much of the externalities that buyers and sellers impose on each other. To illustrate this, consider a hypothetical extreme in which the network effect parameters e_b and e_s approach their maximum value of 1. In this case, the gatekeeper's motivation for self-preferencing (in Corollary 5) perfectly overlaps with the surplus of both buyers and sellers (in Corollary 6), and thus becomes fully aligned with social welfare. Strong network effects, in this sense, naturally regulate the gatekeeper's behavior by incentivizing it to account for the welfare of both sides of the market.

This stands in contrast to an alternative model in which sellers—who select and pay for ancillary services—are given direct control over whether the 1P or 3P service is promoted or supported. In such a setting, sellers are likely to favor the service that offers them higher quality. However, because they internalize only part of the value that buyers derive from these services, they may prioritize their own profits over overall service quality for both parties. As a result, they can promote services that are more lucrative for them but less desirable to buyers. When cross-group network effects are strong, this misalignment can significantly reduce buyer participation, thereby amplifying welfare losses. Therefore, in digital markets characterized by strong cross-group network effects, centralized self-preferencing by a gatekeeper may lead to more efficient outcomes than a fragmented, seller-driven selection process.

Pricing Instruments

Next, we explore the possibility that the gatekeeper firm can charge both buyers and sellers. In this context, a buyer's utility when her interaction is fulfilled by ancillary service i is denoted as $U_b = v_i^b - m_B$, where m_B is the gatekeeper's charge from buyers in the primary market. Accordingly, firm G 's profit becomes $\pi_G = m_B n_B + m_S n_S + p_G Q_G$. All other setups remain the same as in the benchmark.

As illustrated in the Appendix, when the gatekeeper charges both buyers and sellers, the impact of self-preferencing on user participation and their surplus remains the same as in the benchmark. As for the gatekeeper's motivation for self-preferencing, it changes as shown in the following corollary.

Corollary 8. *If firm G charges both buyers and sellers in the primary market, there exists a cutoff $\bar{\beta}_s$ such that the practice of self-preferencing is profitable for firm G when $\beta_s > \bar{\beta}_s$, and non-profitable otherwise. Precisely, $\bar{\beta}_s \leq -2t$ when $\beta_b \geq 0$ and $\bar{\beta}_s > -2t$ otherwise.*

When the gatekeeper charges both buyers and sellers, its strategy regarding self-preferencing is jointly determined by both buyers' and sellers' preferences over ancillary services. When buyers prefer 3P sellers over 1P sellers, the motivation for self-preferencing diminishes, i.e., $\bar{\beta}_s > -2t$. However, when buyers prefer 1P sellers over 3P sellers, firm G is more likely to engage in self-preferencing than in the benchmark, i.e., $\bar{\beta}_s < -2t$. In this context, the welfare implications following self-preferencing become more flexible: both buyers and sellers can either be better off or worse off when self-preferencing is profitable for the gatekeeper. These findings can be summarized in Corollary 9.

Corollary 9. *In a two-sided market where firm G charges both buyers and sellers for the primary product, if firm G finds self-preferencing profitable, then both buyers' and sellers' surplus can either increase or decrease following self-preferencing. Precisely,*

- I) if firm G finds self-preferencing profitable and buyers prefer 3P sellers over 1P sellers ($\beta_b < 0$), then self-preferencing always increases sellers' surplus while decreasing buyers' surplus;*
- II) if firm G finds self-preferencing profitable and buyers prefer 1P sellers over 3P sellers ($\beta_b > 0$), then self-preferencing*
 - II-a) increases both sellers' and buyers' surplus when $\beta_s > -2t$, and*
 - II-b) increases buyers' surplus while decreasing sellers' surplus when $\beta_s < -2t$.*

Comparing Corollary 9 with Corollary 4, we find that the distinction arises when $\beta_b > 0$ and $\beta_s < -2t$. Under these conditions, self-preferencing increases buyer surplus but reduces seller surplus when the gatekeeper charges both sides. In contrast, in the benchmark, self-preferencing increases surplus for both buyers and sellers. This discrepancy stems from the gatekeeper's stronger incentive to self-preference when buyers contribute to its profits and exhibit a strong preference for the 1P ancillary service. In such cases, the gatekeeper is motivated to engage in self-preferencing even at the expense of seller welfare.

Long Run Impact

Building on the previous findings, we now examine the necessity of regulating self-preferencing over the long term. So far, our focus has been on the short-run effects, assuming a fixed number of firms in the adjacent market. This approach aligns with the narrow time frame of our FCA case analysis, which precedes January 2023, when Google was accused of illegally monopolizing advertising technology in

the U.S.¹⁷ Since the digital advertising market is highly dynamic, such cases can influence Google’s global strategy, making it difficult to empirically observe the long-run effects of the FCA regulation.

However, while our focus has been on short-run impacts, it is important to consider whether these practices could harm the market in the long run and whether regulatory intervention is necessary. The answer is affirmative for several reasons. First, as shown in the previous analysis, the practice of self-preferencing reduces the profits of competing firms. Over time, these profit reductions erode competitors’ incentives to innovate, which ultimately harms economic growth.

Beyond these negative effects, even if we focus solely on the market without considering innovation, regulating self-preferencing remains necessary. Although the FCA remedy resulted in a reduction in social welfare as shown in the introduction, this does not imply that regulating self-preferencing is inherently harmful to welfare. As demonstrated in the next section, alternative self-preferencing regulations, when carefully designed, can indeed improve social welfare.

Furthermore, suppose the competition authority refrains from monitoring self-preferencing practices, allowing Google to monopolize the ancillary service market through self-preferencing with no regulatory pressure. In this context, we must assess whether permitting the monopolization of the ancillary market in this hypothetical scenario would lead to lower social welfare compared to a competitive ancillary service market under regulatory pressure. This can be illustrated by a simple scenario where $\beta_b = 0$, which represents a market where buyers view all sellers and ancillary services as identical.

In this context, we compare the market equilibrium before and after the gatekeeper successfully monopolizes the adjacent market, as detailed in the Appendix. We find that once the gatekeeper monopolizes the ancillary service market, sellers are likely to earn a lower surplus and are discouraged from entering. Buyers, in the absence of network effects, experience no change in surplus in this example. However, when positive network effects are accounted for, the reduction in seller participation may negatively affect buyers as well, ultimately diminishing buyer surplus. The results highlight that while regulating self-preferencing is challenging and unintended oversights in remedy design may generate welfare losses in the short run, such market monitoring and regulation are both necessary and highly needed in the long run.

¹⁷See <https://www.justice.gov/archives/opa/pr/justice-department-sues-google-monopolizing-digital-advertising-technologies>

4 Empirical Illustration

We now revisit the FCA case and the evidence in Figures 1 to make two contributions. First, we provide a concise but informative ex post evaluation of the FCA intervention. Given the central role of the adtech market in financing digital platforms, it is crucial to develop an evidence-based understanding of how regulation affects market outcomes. This is especially timely as other antitrust authorities, notably in the European Union and the United States, are addressing the alleged abuses of Google in this sector. Second, we show how our model can be used to empirically analyze alternative counterfactual scenarios, with a focus on evaluating different regulatory remedies against self-preferencing.

FCA-Google AdTech Remedy Discussion

The empirical analysis focuses on two key outcomes: publishers' advertising revenue and the number of unique advertisers placing ads on each domain. These metrics correspond to the FCA's theory of harm, which argued that Google's self-preferencing practices misled publishers about bid prices and, in turn, discouraged advertiser participation. More importantly, both metrics can be viewed as indicators of welfare. Intuitively, if the intervention improves advertisers' profitability, their entry should increase. Similarly, for publishers, if the intervention improves bid quality and presents them with higher bids, they should be able to obtain higher revenue from selling their advertising slots.

Further details on the institutional background of the adtech industry, as well as the FCA case, are provided in the Appendix. In brief, the FCA fined Google in June 2021 for favoring its own programmatic advertising services. The investigation focused on the two mandatory and complementary adtech layers used by publishers—ad servers and Supply-Side Platforms (SSPs)—and concluded that Google leveraged its dominant ad server to favor its own SSP, disadvantaging rivals. To restore competition in France, Google agreed to implement five commitments, with their implementation deadlines spread between 2011 and 2022, as detailed in the Appendix. We focus on the last commitment implemented in June 2022: granting rival SSPs the same access to ad auction data as Google's own services.

There are several reasons for our focus on this data-sharing commitment. First, the data-sharing commitment was critical because SSPs rely heavily on auction data to optimize bidding strategies. Google's dominance in the ad auction market was largely sustained by its superior access to such data, providing it with significant informational advantages. By contrast, competing SSPs often lacked the information to understand lost auctions and improve their performance. Second, all other commitments were addressed prior to the implementation of this data-sharing commitment, with a

relatively concentrated timeline in 2021. To isolate the effect of the data-sharing commitment from other commitments, we are analyzing a time window in 2022, during which all other unfair practices had been generally removed and only the data-sharing commitment was implemented.

If the FCA commitment had effectively restored competition, we would expect both advertiser participation and publisher revenue to increase. However, as shown in Table 2, the simple comparison of averages pre- and post-intervention indicate a decline in France that is sharper than in other comparable countries, suggesting that the remedy may have produced unintended consequences.

Table 2: Advertiser Count and Revenue Following FCA Regulation

		Top 50 Popular Domains		Top 50 Popular News		Top 10 Popular News	
		France	Others	France	Others	France	Others
Advertiser Count	Before	626.18	413.75	448.23	206.26	861.76	259.76
	After	311.19	321.42	197.28	148.46	342.00	207.47
Revenue	Before	270.14	454.81	99.45	92.19	340.19	247.94
	After	115.89	406.04	55.13	79.37	117.19	206.09

Notes: The first two columns consist of the top 50 web domains in EEA and OECD country. The third and fourth columns consist of top 50 popular domains in news and media of 10 European countries, and the last two columns restricted the new and media domains to top 10. The time frame is between January 2022 and January 2023 and the sampling details are provided in the Appendix.

The descriptive evidence in Table 2 is supported by more sophisticated empirical strategies. In particular, Figure 1, introduced earlier, applies a synthetic control approach following Abadie et al. (2010). Prior to the intervention, France and the synthetic control group exhibit similar trends; however, they diverge sharply afterward. Unexpectedly, both publisher revenue and advertiser participation decline in France following the data-sharing requirement. As shown in the Appendix, the findings are robust to alternative specifications, different models, and a different sample of popular news and media websites in the EEA.

Translating these observations into our model, both publishers (sellers) and advertisers (buyers) experience a reduced surplus following self-preferencing alleviation. As demonstrated in Corollary 4, this outcome, i.e., both buyers and sellers benefit from self-preferencing, arises when advertisers derive limited value from interacting with publishers through 3P services (non-Google SSPs).¹⁸ In this context, while reducing self-preferencing on the publisher side may improve the performance of competing ancillary services, this benefit can be offset when the gatekeeper's adtech dominates

¹⁸The statement remains robust when the gatekeeper firm charges both sides of the market, as shown in Corollary 9, and when self-preferencing practices simultaneously affect transaction value, as discussed in the Appendix regarding alternative self-preferencing definition.

the advertiser side and limits the value advertisers derive from interacting with publishers using rival adtechs. As a consequence, regulating self-preferencing solely on the publisher side—without simultaneously enhancing the value of competing ancillary services on the advertiser side—may push publishers toward adtech choices disliked by advertisers. This, in turn, discourages advertiser participation, reduces the value publishers receive from engaging in the market, and ultimately fails to improve overall welfare.

A natural follow-up question is: are the insights drawn from our theoretical model supported by market evidence? Valuable information can be gained from investigations in the European Union and the U.S. In parallel with the FCA, both the DoJ and the EC have been investigating Google's self-preferencing practices in the adtech industry.¹⁹ Notably, unlike the FCA, both the EC and DoJ have extended their investigations to include the advertiser side of the market.

Their findings concerning the advertiser side in the adtech industry can be summarized as follows: (i) the majority of advertisers rely on Google to purchase ad slots from publishers,²⁰ and Google's dominance fosters strong network effects, further increasing the scale and diversity of its advertiser user base; (2) Google's adtech stands apart from competitors due to its data-targeting advantages, powered by a vast range of user data collected from its leading products, such as Chrome, Gmail, and Google Search, as well as data from its ad networks on the publisher side; (3) Google's ad-buying tools typically restrict the performance of non-Google adtech used by publishers, so that most advertisers tend to derive a low value and face a diminished success rate of completing transactions through non-Google SSPs.

As pointed out by the DoJ regarding the latter issue, advertisers using Google Ads are prohibited from submitting bids to competing SSPs based on the same bid modeling and targeting data used for Google's own SSP. In July 2017, Google launched "Project Poirot," which artificially and systematically reduced the bids sent to rival SSPs. During this period, the default setting of Google's ad-buying tool, Display & Video 360 (DV360), was updated to automatically opt into Project Poirot. Due to the default effect, over 99% of advertising campaigns were subjected to this practice. Consequently, the performance of interactions between advertisers' ad-buying tools and non-Google SSPs is likely to be poor, with a low probability of resulting in successful transactions. In other words, advertisers (buyers) in the current display advertising market are likely to derive low value

¹⁹See <https://www.justice.gov/opa/pr/justice-department-sues-google-monopolizing-digital-advertising-technologies> and https://ec.europa.eu/commission/presscorner/detail/en/ip_23_3207 for details.

²⁰The market share of Google Ads has exceeded 90% in both the US and Europe. Such a high market concentration has been achieved through Google's varied strategies in past years. One key factor influencing advertisers' choice of ad-buying tools is the access to valuable advertising inventory. Google has effectively bundled several of its critical advertising inventory, such as Google Search, YouTube, Gmail, and the Google Play Store, with its technology. For example, advertisers seeking to purchase advertising on YouTube are typically required to use Google's ad-buying tools, leaving them with no alternative but to adopt Google's technology.

from interacting with publishers (sellers) through 3P services (non-Google SSPs). Such findings are perfectly consistent with our theoretical insights for the observed empirical patterns in France.

Our model also predicts that Google is motivated to adjust its price following the FCA regulation. Although we did not find any announcement regarding price changes specifically in the French market during the period of our study, this does not imply that there were no changes in adtech prices. As highlighted in existing research, Google possesses significant flexibility in adjusting its pricing strategy within the display advertising market without drawing attention (Hoppner et al., 2021). The pricing of adtech is often viewed as a “black box” for publishers and advertisers. Most publishers and advertisers only receive bills from Google without understanding how the company charges each transaction. For example, in the SSP adtech layer, the charge is on an individual basis for each impression and is highly dynamic. As long as the “average” revenue share remains at the contractually set rate, Google can essentially set its adtech charge at any level on a case-by-case basis. This dynamic pricing mechanism and its associated rules allow Google to adjust its adtech charge for each transaction, granting it considerable freedom in the pricing strategy. In this market context, Google can adjust its pricing strategy within a wide range without making announcements.

Counterfactual Policy Comparison

We next combine the model with display advertising data to evaluate counterfactual policy interventions. Our study focuses on the welfare dimension for several key reasons. First, welfare is one of the most critical metrics valued by both society and policymakers, as it reflects the overall well-being of consumers and sellers, guiding regulatory decisions that promote fairness and efficiency in the market. Second, as shown in the previous analysis, user surplus is positively correlated with market participation, meaning that analyzing welfare can provide valuable insights into market growth and user engagement. Third, although other measures—such as prices or profits—are also important, they often present challenges in terms of data availability and reliability. For instance, as discussed earlier, the pricing of adtech is not transparent. Google can set its adtech charges on a case-by-case basis and adjust prices without publicly altering its pricing rules, making it extremely difficult to detect price fluctuations.

The primary goal of our empirical illustration below is to calibrate the theoretical framework, rather than to provide a full-fledged structural representation of the adtech market. We aim to organize the counterfactual analysis by mapping empirically relevant parameters to key theoretical mechanisms, rather than reproducing every institutional detail. While the quantitative estimates may vary depending on assumptions or constraints, the qualitative conclusions remain robust. Additionally,

beyond the empirical illustration, we also map the two alternative remedies to our theoretical model and analyze their impact through comparative statics in the Appendix. The theoretical analysis yields qualitatively similar conclusions.

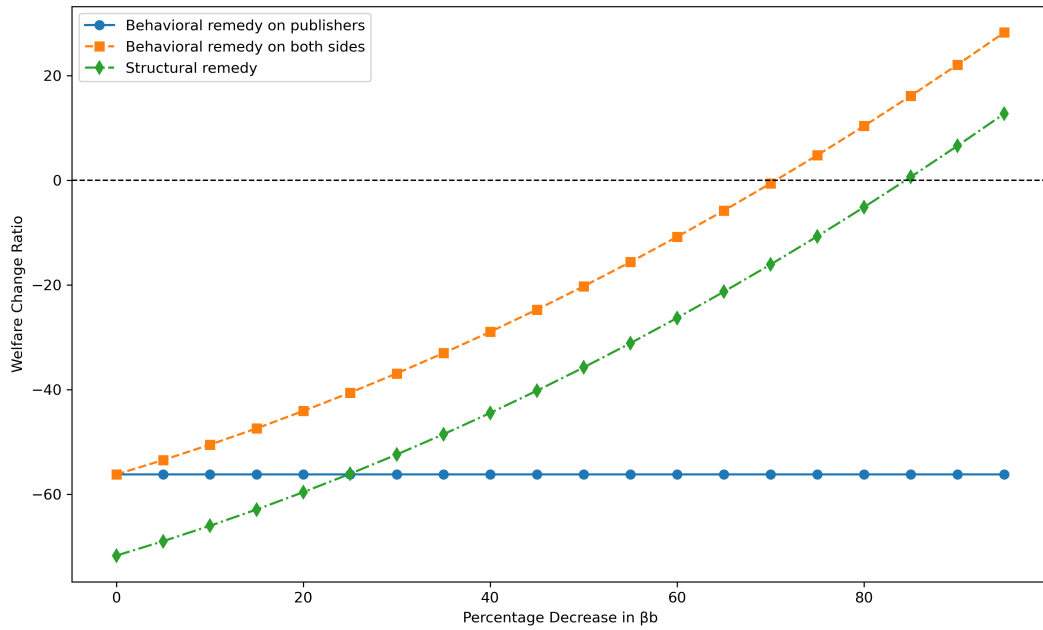
Specifically, we first estimate the welfare loss among the top popular French websites in news and media after the FCA regulation. We then conduct two counterfactual exercises to evaluate alternative remedies. The first exercise considers an enhanced behavioral intervention that aims to increase the value of 3P services for both publishers and advertisers. The second exercise examines whether, and under what conditions, a structural remedy could yield more effective welfare improvements than behavioral remedies. This structural approach is motivated by Google’s recent proposal to divest AdX as part of its settlement negotiations with the EC. Compared to behavioral remedies, divestment is expected to be more effective in removing the gatekeeper’s incentive for self-preferencing. However, given the complexity of digital markets and the coordinating role of Google, its ultimate effect on welfare remains uncertain.

We begin the analysis by estimating model parameters using the Method of Simulated Moments (MSM). Precisely, we match model-generated moments of advertiser count and publisher revenue with their empirical counterparts from the display advertising market, focusing on the top 100 most popular news and media websites. Details of this estimation are provided in the Appendix, with all the estimated parameters satisfying our model assumptions.

Based on the estimation results, we simulate the impact of the data-sharing remedy implemented by the FCA. This remedy improved competing adtech options for publishers, while Google remained the dominant choice for advertisers and was not restricted from limiting advertisers’ value over interacting with non-Google adtech used by publishers. On average, our estimation suggests that advertisers derive significantly less value from interacting with publishers using third-party alternatives compared to those using Google adtech. On the publisher side, the value of the 3P service was also lower than that of 1P services before the FCA regulation. After the regulation, over 90% of this quality gap disappeared for publishers. However, as the theoretical model predicts, addressing self-preferencing asymmetrically—solely on the publisher side, without addressing the advertiser side—reduces surplus for both advertisers and publishers. Consistent with this prediction, we observe that, rather than improving, the change in total social surplus—defined as the combined surplus of advertisers and publishers—turned negative, with a magnitude of around 56.24%. The welfare loss level is denoted by the blue line in Figure 4.²¹The estimates are generally consistent with empirical observations in

²¹The FCA remedy does not address the advertiser side or induce any changes in β_b ; that is, the percentage decrease in β_b , namely, the x coordinate, is already zero. However, to facilitate comparison between the FCA remedy and alternative remedies, we plot the FCA welfare loss as a horizontal line rather than a point in Figure 4.

Figure 4: Social Welfare Impact of Different Regulations



Note: The blue line shows the change in total welfare in France following the FCA regulation; the orange line reflects an alternative remedy improving 3P services for both publishers and advertisers; and the green line represents a divestment scenario. The horizontal axis indicates the extent of reduced vertical differentiation for advertisers, with 100% meaning no perceived vertical differentiation.

the French display advertising market, where the advertiser count dropped to nearly half after the regulation.

Such a reduction in social welfare can be addressed with an alternative remedy design. Specifically, we investigate the potential impact of an enhanced remedy: instead of addressing self-preferencing only on the publisher side, this alternative policy also targets Google's current strategy that discourages advertisers from interacting with publishers using non-Google adtech. In Figure 4, the orange line shows the percentage change in total welfare in France when, in addition to the same magnitude of rival adtech improvement for publishers as under the FCA regulation, a behavioral remedy also reduces the relative advantage of 1P over 3P services for advertisers. The horizontal axis represents the extent to which vertical differentiation between 1P and 3P services is reduced. When this ratio reaches 100 percent, all vertical differentiation disappears, and advertisers perceive no difference between interacting with publishers through 1P or 3P services.

Our results show that while the FCA remedy alone leads to a decline in total welfare, its effectiveness improves substantially when it also enhances the value of 3P services for advertisers.

Specifically, we find that greater reductions in vertical differentiation lead to increasingly larger gains in total welfare. When the vertical differentiation between 1P and 3P services for advertisers is reduced, with a magnitude of over 70.54%, the overall welfare effect of the regulation becomes positive.

Next, we evaluate the potential effects of a structural remedy. During negotiations between the EC and Google, the possibility of divesting AdX was discussed. To disentangle the underlying forces, we decompose the overall changes in market outcomes after a potential divestment as follows:

$$\begin{aligned}
\Delta EQ &= EQ'(\beta'_s, \beta'_b) - EQ(\beta_s, \beta_b), \\
&= EQ'(\beta'_s, \beta'_b) - EQ(\beta'_s, \beta'_b) + EQ(\beta'_s, \beta'_b) - EQ(\beta_s, \beta_b), \\
&\approx \underbrace{EQ'(\beta'_s, \beta'_b) - EQ(\beta'_s, \beta'_b)}_{\text{induced by market structure adjustment}} + \underbrace{\frac{\partial EQ(\beta_s, \beta_b)}{\partial \beta_b}(\beta'_b - \beta_b)}_{\text{induced by } \beta_b \text{ adjustment}} + \underbrace{\frac{\partial EQ(\beta_s, \beta_b)}{\partial \beta_s}(\beta'_s - \beta_s)}_{\text{induced by } \beta_s \text{ adjustment}},
\end{aligned} \tag{14}$$

where ΔEQ represents the overall changes in market outcome, EQ is the equilibrium strategy before divestment, EQ' is the equilibrium strategy after divestment, β_s and β_b measure the vertical differentiation of ancillary services for sellers and buyers before the intervention, and β'_s and β'_b measure the vertical differentiation after the intervention.

As shown in Equation 14, the overall market change following divestment can be categorized into three effects. First, given the ancillary service values, the equilibrium strategy changes due to modifications in market structure. Second, given the market structure and equilibrium strategy, buyers' value of 3P services may improve as self-preferencing incentives are completely removed after the divestment. Third, similar to the buyer side, sellers' value of 3P services may also improve following the divestment. The third effect, driven by an improvement in sellers' value of 3P services, is similar to the FCA intervention and has been well-studied in previous analyses. Therefore, the focus of our analysis is the first two new effects and their overall impact on social welfare.

To investigate these effects, we analyze a scenario where the ownership of the 1P service transfers from Google, the gatekeeper firm, to an independent company. After divestment, the gatekeeper derives profit solely from the primary market and adjusts its strategy accordingly. By solving the profit maximization problem, we can easily derive the new market equilibrium following divestment, as shown in the Appendix. Based on this equilibrium, we then simulate market outcomes and compute changes in total welfare. In Figure 4, the green line shows the total welfare change ratio in France after the divestment, with the horizontal axis representing the reduction in vertical differentiation between 1P and 3P services for advertisers.

Interestingly, we find that divestment is generally less effective than a behavioral remedy when both

interventions enhance the value of 3P services for both sides of the market by the same amount. In other words, the first effect in Equation 14 is always negative. Specifically, if divestment improves the value of 3P services only for publishers without affecting advertisers, the expected reduction in total welfare is approximately 71.75%. This indicates that divestment is even less effective than the FCA remedy currently in place. This outcome occurs because divestment lowers the profit that each user generates for the gatekeeper firm. Consequently, the dominant firm's incentive to expand market size diminishes, leading to a lower aggregate demand for market participation. At the same time, competing firms tend to experience reduced profits after divestment. Over the long run, this negative impact may accelerate their exit, ultimately decreasing overall competition in the ancillary service market.

Nevertheless, our results suggest the second effect in Equation 14 is positive, meaning that the effectiveness of divestment increases when the 3P service value improves for advertisers. The finding underscores that divestment can outperform behavioral remedies in enhancing social welfare, provided it more effectively removes Google's incentive to favor its own services on both sides of the market. In other words, if divestment results in a substantially greater improvement in advertisers' value of 3P services compared to behavioral interventions, its impact on welfare becomes more favorable. Specifically, when the vertical differentiation between the 1P and 3P services is reduced by the same magnitude as under the FCA remedy for publishers and by more than 84.44% for advertisers, the total social welfare increases as a result of divestment.

In summary, compared to the FCA regulation whose impacts are largely captured by the third component of Equation 14, divestment generates two additional and conflicting effects on social welfare. On one hand, divestment reduces the gatekeeper's incentive to maintain a large market and may discourage entry, thereby shrinking overall participation. On the other hand, it can more effectively eliminate the gatekeeper's motivation for self-preferencing, potentially leading to a stronger improvement in advertisers' value of 3P services. As a result, there is no definitive conclusion about which regulatory approach is superior.

Our findings offer a clear policy insight: structural remedies like divestment can be highly effective, but only if they sufficiently improve the quality of competitive ancillary services across the entire market chain for both sides of the market. When applied to just one side of the market, such remedies risk repeating the limitations of behavioral interventions, as seen in the French market following the FCA's action. Effective regulation must account for the bilateral nature of digital platforms and ensure that interventions restore functionality for both buyers and sellers. Otherwise, policies aimed at promoting competition may unintentionally fragment the market and reduce overall efficiency.

To check the robustness of our conclusions, we provide the following analysis in the appendix. First,

we examine the welfare change across three remedies while relaxing estimation constraints or changing assumptions. Specifically, we check how the results change when we relax the estimation constraints that require simulated advertiser counts, publisher revenue, and the estimated self-preferencing measure to decrease after regulation. Without these constraints, the qualitative changes in the simulated and empirical moments following regulation do not align, and the magnitude of welfare changes varies accordingly. Additionally, we adjust the normalization assumption of horizontal differentiation. We find that greater horizontal differentiation reduces the magnitude of the welfare change, as it leads to more competition in the ancillary services market, which in turn mitigates the effects of self-preferencing regulation.²² Despite these modifications, the overall qualitative conclusions remain consistent.

5 Conclusions

The FCA’s 2021 case against Google highlights the complex challenges regulators face in addressing self-preferencing in digital platform markets. The behavioral remedy imposed—requiring Google to share auction data with rival SSPs—was intended to level the playing field but led to unintended consequences: both publisher revenue and advertiser participation declined. This case illustrates how traditional remedies, even when well-intended, can produce adverse effects in complicated digital markets.

To better understand these counterintuitive outcomes, we developed a unified framework capturing core features of cross-market self-preferencing: sellers choose and pay for ancillary services, yet their choices influence outcomes for buyers as well. We show that a dominant gatekeeper can profit from self-preferencing even when its own ancillary service is not superior in quality. More importantly, the welfare implications can be either positive or negative, depending on buyers’ and sellers’ preferences and the platform’s pricing instruments.

These theoretical insights reveal two critical lessons. First, remedies that target only one side of the market—without accounting for cross-side dynamics—can misalign incentives and reduce welfare. By addressing self-preferencing solely on the publisher side, the FCA intervention pushed publishers toward rival technologies that advertisers were less willing to use, diminishing participation on both sides. Second, curbing self-preferencing may also reduce the gatekeeper’s incentive to maintain a large, well-functioning ecosystem. This mechanism is not just theoretical: Apple’s recent response to

²²In a previously circulated version of this paper, we present the estimation results when estimation constraints are not considered and horizontal differentiation set at $t = 100$.

an EC fine for restricting communication of alternative payment options in its App Store exemplifies it. Although Apple formally lifted the marketing restriction, it introduced two new fees—a 5% user acquisition fee and a 10% services fee on out-of-store transactions. This illustrates how gatekeepers may re-optimize their pricing strategies post-regulation and diminish market size.

Our paper resonates with ongoing policy debates and yields several implications. First, the EC’s draft guidelines on Article 102 of the Treaty on the Functioning of the European Union (TFEU) propose, for the first time, a structured definition of self-preferencing as an exclusionary abuse.²³ Our findings underscore that regulation must account for the two-sided nature of self-preferencing, even when the ancillary service is not a bottleneck input. Second, the recent ruling in the US against Google signals a growing willingness to consider structural remedies. Our model supports this approach, but only in cases where structural changes improve rival quality for both sides of the market. Indeed, through counterfactual simulations, we identify conditions under which alternative behavioral or structural interventions improve welfare, offering actionable thresholds for policy design.

Although the paper focuses on Google’s self-preferencing in the adtech case, its model and findings are applicable to other antitrust cases. We conclude by briefly discussing one such application: Amazon’s self-preferencing practices in the e-commerce and fulfillment service markets. Specifically, the EC has accused Amazon of promoting sellers who use its own logistics services, giving them unique advantages on its platform. In this context, Amazon leverages its dominant position in the e-commerce market—the primary product market—to favor its own logistics services, which are complementary ancillary services that sellers rely on to complete transactions with buyers. This self-preferencing practice aligns with our definition, as sellers using rival logistics services are less likely to be recommended by Amazon and thus obtain lower surplus in the market.

According to our findings, while it is important for regulators to ensure fair treatment of all sellers on Amazon’s platform, it is also critical for policymakers to understand buyers’ preferences for Amazon’s fulfillment services and how competing firms like FedEx or UPS are likely to evolve after such regulation. If self-preferencing regulation not only guarantees fair treatment of sellers but also encourages rival logistics companies to improve their service quality for buyers, the interventions are likely to be effective in promoting social welfare. However, if self-preferencing regulation reduces competition and motivates rival shipping companies to decrease their service quality or raise their prices, we must be cautious about the risk of unintended regulatory backfire in the market.

Our study suggests several avenues for future research. One is to extend the framework to multiple forms of self-preferencing, such as Amazon’s simultaneous promotion of its own brand and logistics

²³See <https://eur-lex.europa.eu/eli/treaty/tfeu.2012/oj/eng>.

services. Another is to empirically examine how regulation affects multihoming and longer-term market dynamics beyond short-run surplus effects. Lastly, introducing competition in the primary market may reveal additional forces shaping self-preferencing incentives.

In summary, this paper provides the first integrated theoretical and empirical analysis of cross-market self-preferencing in digital platform ecosystems. As public scrutiny of digital platforms intensifies on both sides of the Atlantic, designing regulatory evidence-based interventions that are both targeted and comprehensive will be essential for improving market outcomes.

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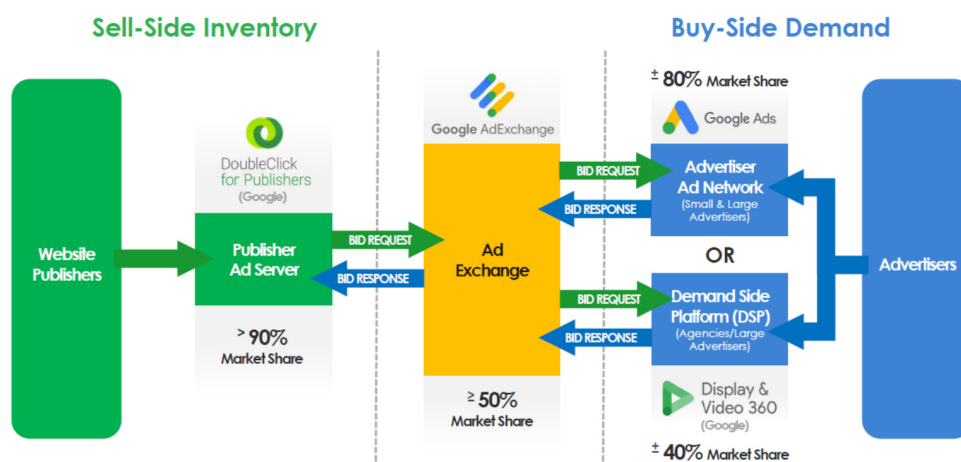
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A Appendix

A.1 Institutional Background on the Adtech Industry

In the display advertising market, publishers, like websites and apps, constitute the supply (seller) side that offers available ad spaces, and advertisers represent the demand (buyer) side that purchases spaces to catch eyeballs. Both publishers and advertisers rely on an intricate chain of advertising technologies to accomplish interactions with each other.

Figure A.1: DOJ's Diagram of Google's Ad Business



In a simplified adtech stack as shown in Figure A.1, there are three main blocks of adtechs. Precisely, ad servers work as the first technology layer for publishers. The servers assist publishers in managing ad inventory and displaying relevant ads. The second technology layer for publishers encompasses supply-side platforms (SSPs), which actively solicit bids from advertisers and organize auctions for specific impressions. Advertisers, likewise, need ad-buying tools to optimize bids and participate in auctions organized by SSPs. Large advertisers typically use demand-side platforms (DSPs) that offer sophisticated and customizable ad-purchasing tools. Smaller advertisers, on the other hand, often rely on simpler advertising solutions provided by ad networks.

Each layer in the adtech stack plays an indispensable role and interacts in a sequential manner as follows: when a user visits a website or app, the publisher's ad server first receives information about available impressions, i.e., ad slots for that particular user. The server then requests bids for these impressions from the connected SSPs. The SSPs transmit these requests to advertisers, who then use ad-buying tools to determine the bid amount on their behalf. The ad-buying tools then send the winning

bids back to the SSPs. After receiving bids from multiple ad-buying tools, the SSPs conduct an auction among all the bids, choose the winner, and return the winning bid to the publisher's ad server. Lastly, the publisher's ad server conducts a final auction among the bids received from its SSPs and notifies the selected ad to be displayed.

Among these three components, the publisher ad server and SSP markets have been the focus of the FCA investigation. The case is centered exclusively on the adtechs used by publishers, without considering the advertiser side. Compared to other layers of the adtech stack, publisher ad servers tend to exhibit a high degree of "stickiness": publishers rarely switch ad servers due to the substantial costs and risks involved. As the former CEO of DoubleClick pointed out in a Google internal strategy meeting²⁴:

"My view is nothing really matters but the platform [publisher ad server]. Nothing has such high switching costs... Switching platforms is a nightmare. Takes an act of God to do it."

In the current display advertising market, Google is active in all layers of adtech stack. The company provides an ad server, "DoubleClick for Publishers (DFP)", and an SSP, "Google AdExchange (AdX)" for publishers. The DFP and AdX both fall under Google's re-branded "Google Ad Manager" services, but they fulfill distinct functions. On the advertiser side, Google offers a DSP, "Display & Video 360 (DV360)", for large advertisers and an ad network, "Google Ads", for small advertisers. More importantly, Google holds dominant positions in multiple adtech layers. In the United States, for instance, Google's DFP controls over 90% of the publisher ad server market. In terms of SSPs, Google manages more than 50% of ad sales. On the advertiser side, Google Ads captures around 80% of demand, while DV360 holds about 40%. Similar trends are observed in Europe.²⁵

In June 2021, the FCA concluded its investigation and fined Google €220 million for prioritizing its programmatic advertising services under the Google Ad Manager brand.²⁶ According to the FCA, Google had been engaging in reciprocal self-preferencing across these two adtech layers: it provided more advantageous terms to its SSP (AdX) compared to competing firms, including allowing AdX to optimize its algorithm using competitors' bids and restricting competitors' data access to auction outcomes. Furthermore, Google's AdX was also found to favor its server (DFP) by making it the only ad server fully interoperable with AdX.

As pointed out by the FCA, Google's practices were identified as having substantial

²⁴See <https://www.justice.gov/atr/media/1371646/dl?inline>.

²⁵See https://papers.ssrn.com/sol3/papers.cfm?abstract_id=3929310.

²⁶See <https://www.autoritedelaconurrence.fr/sites/default/files/attachments/2021-07/21-d-11-ven.pdf>.

anti-competitive effects within the markets for publisher ad servers and SSPs. These actions reduced competition among adtech intermediaries and adversely affected both publishers and advertisers. There are signs that publishers utilizing Google’s ad server were not presented with the highest possible bids from competing SSPs. Further evidence, including internal Google documents, also suggests that the company misled publishers about the benefits of its services. Both effects led to revenue loss for publishers.²⁷ For advertisers, Google’s high extraction rates force them to pay inflated prices, leaving them worse off. These market distortions not only harm advertisers and publishers but also undermine the quality of adtech tools and stifle innovation. Ultimately, this situation negatively impacts consumers. Reduced revenue for publishers limits their ability to offer free online content, potentially leading to the introduction of more paywalls or subscription models, thereby restricting consumers’ access to information.

To address the anti-competitive concerns, Google reached an agreement with the FCA to gradually fulfill a series of commitments²⁸ in France, with the full implementation deadline provided in parentheses: (i) allowing fair access to information on the auction process for third-party SSPs (2022 June 30th); (ii) preserving the full contractual freedom of third-party SSPs so that they can negotiate special conditions with publishers and are not blocked from deciding to include or exclude specific buyers (2021 November 15th); (iii) ensuring that AdX no longer uses the price of its competitors in order to optimize its bids in a way that is not reproducible by third-party SSPs (2021 February 15th); (iv) offering guarantees of technical stability, both for third-party SSPs and for publishers (2021 May 15th); and (v) making changes to existing configurations that allow publishers using third-party ad servers to access AdX on-demand in “real-time” (2022 January 15th).

²⁷See <https://awards.concurrences.com/en/awards/2022/business-articles/the-french-competition-authority-fines-a-big-tech-company-eur220-million-for>.

²⁸According to the FCA, the commitments are binding for publishers whose registered place of business is in the European Economic Area (EEA), have material activities in the EEA, and use ad servers or SSPs to serve ads to users with an IP address in France. See https://www.autoritedelaconurrence.fr/sites/default/files/commitments/2021-08/google_commitments_english_version_21d11.pdf.

A.2 Additional Analysis on FCA Case

To investigate whether any clear patterns emerged after the FCA regulation, we collected data from various sources for our main analysis. First, we use Similarweb²⁹, an Israeli firm specializing in web traffic and performance analysis, to identify the top 50 most popular websites across a series of EEA and OECD countries in February 2023. Specifically, our list includes 26 countries: Argentina, Australia, Belgium, Brazil, Canada, Switzerland, Germany, Denmark, Spain, Finland, France, the United Kingdom, Indonesia, India, Italy, Mexico, Malaysia, the Netherlands, Norway, New Zealand, the Philippines, Sweden, Singapore, the United States, Vietnam, and South Africa. Based on this sample list, we then gather advertising performance data from the SEMrush platform.³⁰ The time frame for our sample is from January 2022 to January 2023.³¹ The final sample consists of 23,045 observations.

In the main text, we employ the synthetic control technique proposed by Abadie et al. (2010). In this framework, France is defined as the “treated” group, and our objective is to construct a synthetic control group that best approximates the counterfactual outcomes of France in the absence of the regulation. Specifically, we select potential control countries from the set of EEA and OECD nations based on their pre-treatment outcomes in advertising revenue and advertiser count, aiming to identify the combination of weights on those control countries that most closely resemble France prior to the regulation’s implementation.

Our analysis suggests a sharp decline in both publishers’ advertising revenue and advertisers’ participation in France following Google’s data-sharing commitment. To ensure the validity of our findings, we conducted multiple robustness checks. First, we estimate the impact of regulation using an alternative difference-in-differences identification strategy. Specifically, we utilize the following Two-Way-Fixed-Effects (TWFE) model as our baseline specification:

$$Y_{cdt} = \alpha + \beta(Treated_c \times Post_t) + \lambda_c + \gamma_t + \theta_d + \varepsilon_{cdt}, \quad (15)$$

where Y_{cdt} represents the variable under investigation of domain d in country c during week t . λ_c denotes the country fixed effect, γ_t represents the week fixed effect, θ_d represents the domain fixed effect, and $Treated_c \times Post_t$ is the treatment variable, which is an indicator for the treated group following the policy implementation in June 2022. As shown in Table A.1, we find qualitatively similar

²⁹See <https://www.similarweb.com/>.

³⁰See <https://www.semrush.com/>.

³¹We exclude websites belonging to the FAANG group (Alphabet, Amazon, Apple, Meta, Microsoft, and Netflix), as they may behave differently from other websites.

results to those reported in the main text: a significant decline in both publishers' advertising revenue and advertisers' participation in France following Google's data-sharing commitment.

Table A.1: FCA Remedy Impact on Popular Websites

	(1) Revenue	(2) Revenue	(3) Revenue	(4) Advertisers	(5) Advertisers	(6) Advertisers
Post 06/2022	-40.847** (15.778)			-91.299*** (7.544)		
France	-176.657*** (49.737)			213.912*** (23.781)		
Treated $\times$ Post 06/2022	-114.111 (67.344)	-114.111 (61.623)	-114.111** (43.019)	-223.214*** (32.200)	-223.214*** (29.520)	-223.214*** (16.940)
Week FE	No	Yes	Yes	No	Yes	Yes
Country FE	No	Yes	Yes	No	Yes	Yes
Domain FE	No	No	Yes	No	No	Yes
R-squared	0.003	0.168	0.601	0.012	0.172	0.732
Observations	23045	23045	23045	23045	23045	23045

Our second set of robustness checks uses an alternative sample consisting of the top 100 most popular news and media websites across selected EEA countries, including Belgium, Germany, Denmark, Spain, Finland, France, Italy, the Netherlands, Norway, and Sweden in September 2023.³² As shown in Table A.2, the estimates based on this media sample are broadly consistent with those obtained using the top 50 most popular websites in the EEA and OECD countries.

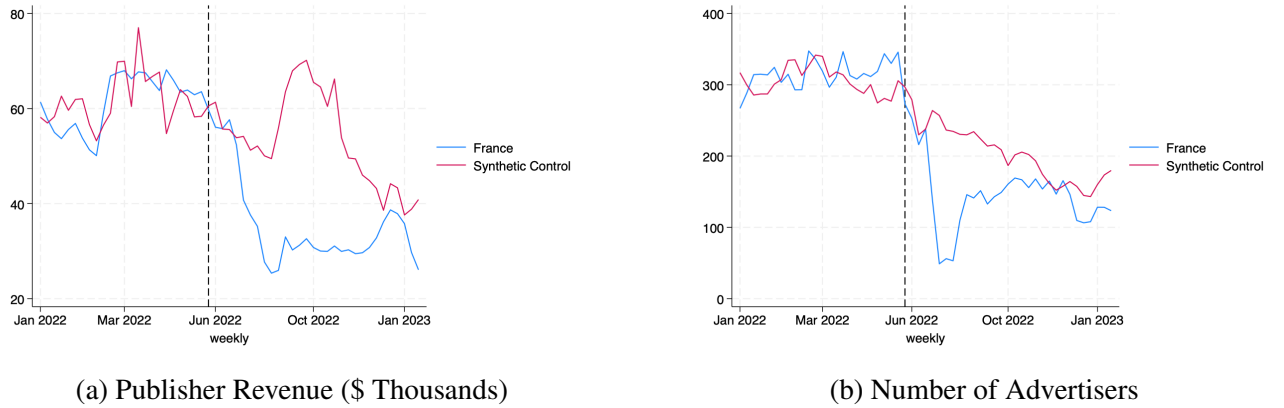
³²Following the same procedure as in the previous analysis, we exclude the FAANG group from the sample, resulting in a final dataset of 30,212 observations.

Table A.2: FCA Remedy Impact on News-and-Media Websites

	(1) Revenue	(2) Revenue	(3) Revenue	(4) Advertisers	(5) Advertisers	(6) Advertisers
Post 06/2022	-8.665** (2.870)			-50.284*** (4.589)		
France	-1.545 (5.540)			123.702*** (8.857)		
Treated $\times$ Post 06/2022	-17.233* (7.688)	-17.233* (7.632)	-17.233*** (4.525)	-119.027*** (12.291)	-119.027*** (11.800)	-119.027*** (7.267)
Week FE	No	Yes	Yes	No	Yes	Yes
Country FE	No	Yes	Yes	No	Yes	Yes
Domain FE	No	No	Yes	No	No	Yes
R-squared	0.001	0.017	0.660	0.014	0.093	0.662
Observations	30212	30212	30212	30212	30212	30212

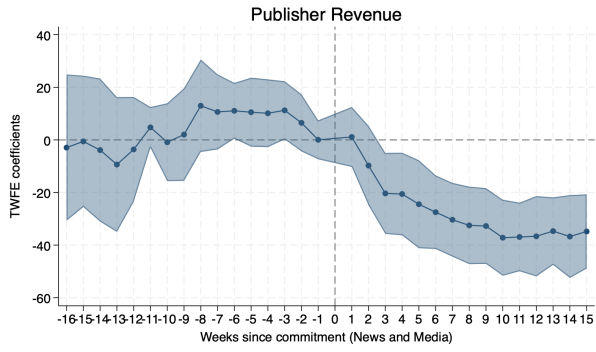
Following the same method, we plot the weekly averages of advertising revenue and advertiser counts for popular news and media domains in both the treated and synthetic control groups in Figures A.2a and A.2b, yielding qualitatively similar results.

Figure A.2: News and Media Market: Pre- and Post-FCA Intervention

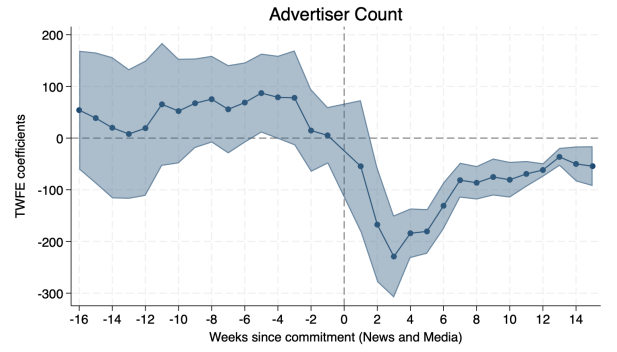


Our third set of robustness checks incorporates time-varying effects using the TWFE model as shown in Figure A.3.

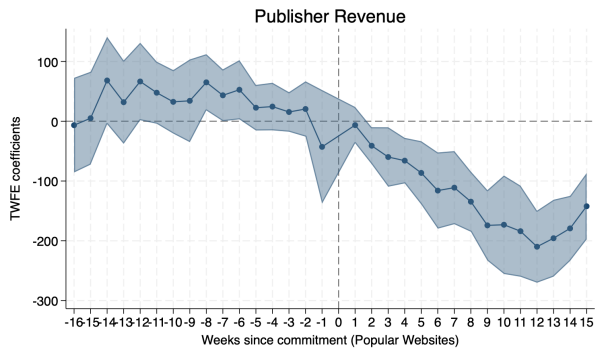
Figure A.3: Time-Varying Effect of the FCA Remedy



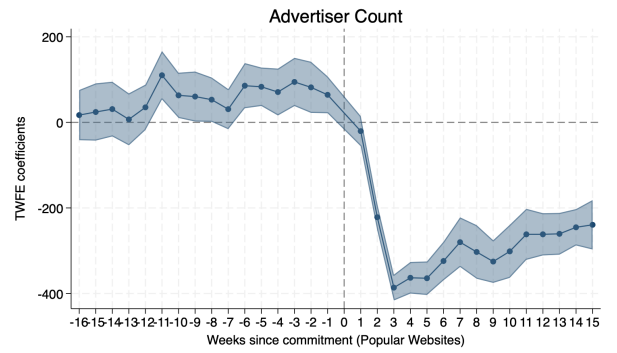
(a) Publisher Revenue (News and Media)



(b) Advertiser Count (News and Media)



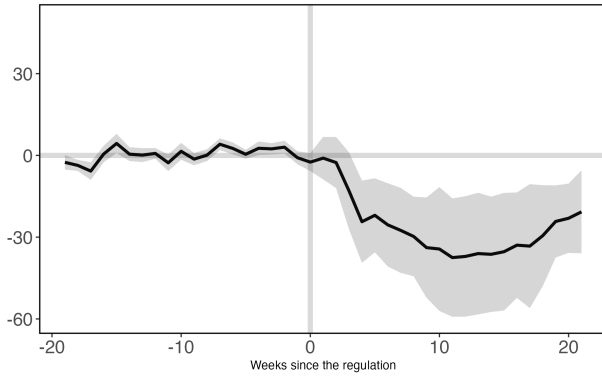
(c) Publisher Revenue (Popular Websites)



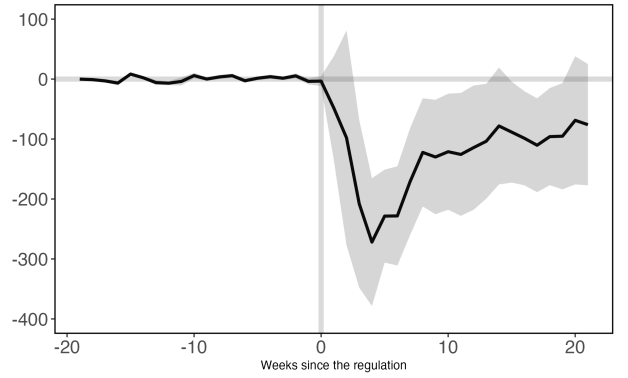
(d) Advertiser Count (Popular Websites)

Figure A.3a and Figure A.3b demonstrate that both publishers' revenue and advertiser participation in news and media websites began to decline following the implementation of the FCA remedy. Furthermore, this downward trend in advertiser participation persisted, while the decline in publisher revenue intensified over time. Similar patterns are also observed on popular websites as shown in Figure A.3c and Figure A.3d. These findings are consistent with our previous results. Importantly, the event study specifications provide evidence supporting our parallel trends assumption, as there are almost no significant differences between the treated and control groups prior to the implementation of the FCA remedy.

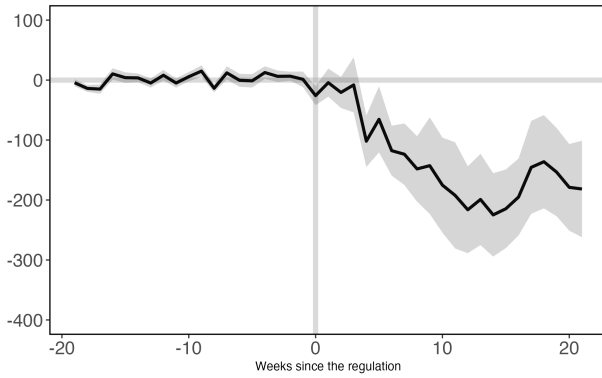
Figure A.4: MCNN Estimation



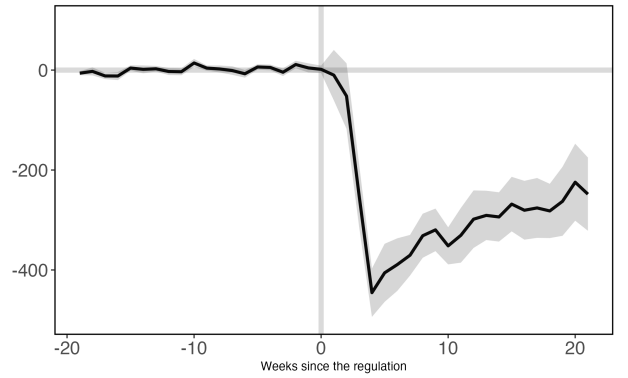
(a) Publisher Revenue (News and Media)



(b) Advertiser Count (News and Media)



(c) Publisher Revenue (Popular Websites)



(d) Advertiser Count (Popular Websites)

Our final set of robustness checks uses the Matrix Completion-Nuclear Norm (MCNN) methodology, as proposed by Athey et al. (2021). Originally developed for forecasting tasks in Computer Science, MCNN leverages machine learning techniques to predict the potential outcomes of treated units. It combines pre-treatment patterns from both control and treated units to impute the counterfactual outcomes of treated units following an intervention, effectively synthesizing a parallel pre-treatment trend. The results are shown in Figure A.4 and generally consistent with our findings.

A.3 Self-Preferencing by Amazon and Apple

Examining the current digital market, it becomes evident that the self-preferencing phenomenon is increasingly prevalent. In addition to Google's self-preferencing in the advertising market, another notable example is Amazon's practice in the e-commerce and logistic markets. In December 2022, Amazon was accused by the EC of discouraging sellers from using rival logistics services to fulfill Amazon orders. When sellers fulfill orders on Amazon, they can either manage storage and delivery independently or outsource these tasks to an operator. Amazon itself offers corresponding services to sellers through its Fulfillment by Amazon (FBA) program. According to the EC's investigation, Amazon has been promoting its own logistics service by providing it with unique advantages on the platform, including (i) non-application of performance metrics to third-party sellers; (ii) obtaining the Prime Badge; (iii) higher probability of being awarded the Buy Box; (iv) participation in special events and offers; and (v) eligibility for "Free Shipping via Amazon."

All these benefits play critical roles in facilitating selling on Amazon. For instance, one significant advantage for FBA sellers is their higher chance of winning the Buy Box. Unlike others, sellers who win the Buy Box can be immediately added to buyers' shopping carts through the "Add to Basket" button or be purchased directly through the "Buy Now" button. Due to this convenience, most Amazon sales occur through Buy Box recommendations. Taking Germany and France as examples, the EC highlighted that between 70% and 90% of total Amazon sales in these countries were finished through the Buy Box between 2017 and 2019. In its communications with sellers, Amazon also stated that "90% or more of sales come from the Buy Box."³³

Another significant advantage of FBA sellers is their increased likelihood of obtaining Amazon Prime labels. Over the past years, there has been steady growth in the number of Prime users on Amazon. These Prime users typically spend more on Amazon than regular users and strongly prefer Prime offers. According to the EC, between 70% and 90% of total spending by Prime users in the German, French, and Spanish Amazon marketplaces was associated with Prime offers. As a consequence, qualifying for Prime labels becomes crucial for sellers to attract demand.

While FBA sellers automatically qualify for the Prime label, sellers who fulfill their own orders (FBM) must meet the stringent requirements of Amazon's Seller Fulfilled Prime (SFP) program. This includes entering into an agreement with an Amazon-designated carrier and complying with shipping terms negotiated by Amazon. These logistical and contractual hurdles create a significant barrier for FBM sellers, effectively nudging them toward using Amazon's fulfillment services.

³³See https://ec.europa.eu/competition/antitrust/cases1/202310/AT_40703_8990760_1533_5.pdf

Given these disparities in treatment, the majority of sellers had little practical choice but to adopt FBA. This widespread adoption not only reinforced Amazon’s dominant position in the e-commerce marketplace but also allowed it to expand rapidly into the shipping and logistics industry. By integrating more sellers into its fulfillment network, Amazon gained comparative advantages over traditional logistics providers, thereby solidifying its market power across multiple layers of the supply chain.

Likewise, in March 2024, the EC fined Apple €1.8 billion for exploiting its dominant position in the App Store to favor its own electronic payment channel. As pointed out by the EC, Apple is the sole provider of the App Store, where app developers and iOS users interact. The company also provides payment technology and channels for app developers to process users’ subscriptions. According to the EC’s investigation, Apple has imposed strong marketing restrictions that prevented streaming app developers from fully informing iOS users about alternative, lower-cost music subscription channels available outside of Apple.

Specifically, these anti-competitive practices include: (i) prohibiting the communication of subscription prices available outside the app to iOS users within the app, (ii) preventing the disclosure of price differences between in-app subscriptions, sold through Apple’s own payment system, and those offered elsewhere, and (iii) forbidding the inclusion of links within the apps that would direct iOS users to the app developer’s website, where alternative subscription options could be purchased. Additionally, app developers were prohibited from contacting newly acquired users—such as via email—to inform them about alternative pricing options after they had set up an account.

As a result, the majority of iOS users complete their subscriptions within Apple’s ecosystem but end up paying significantly higher prices. This is due to the substantial commission fees Apple imposes on developers, which are ultimately passed on to consumers in the form of higher subscription prices. Consequently, iOS users are believed to be severely harmed by Apple’s self-preferencing practices.

A.4 Corollary Proofs

Recall that our benchmark focuses on the meaningful market equilibrium under Assumption 1. Precisely, the cutoffs in Assumption 1 can be expressed as:

$$\begin{aligned}\tilde{v}_G^s &= \frac{6\beta_s t + 45t^2 - \beta_s^2}{36t} \\ \underline{v}_G^b &= \frac{3\beta_b t - \beta_b \beta_s}{6t} \\ \bar{v}_G^b &= \frac{6t + 3\beta_b t - \beta_b \beta_s}{6t} \\ \underline{v}_G^s &= \frac{2\beta_s t + 9t^2 - \beta_s^2}{12t} \\ \bar{v}_G^s &= \frac{12t + 2\beta_s t + 9t^2 - \beta_s^2}{12t}\end{aligned}$$

Proof of Corollary 1 According to Equations (2-3), it is evident that an increase in β_s increases p_G while decreasing p_C . To examine the impact of β_s on the primary product price m_S , we calculate the derivative and obtain:

$$\frac{dm_S}{d\beta_s} = -\frac{\beta_s + 6t}{36t}.$$

Recall that $|\beta_s| < 3t$ under Assumption 1. Thus, the derivative $dm_S/d\beta_s$ is always negative and an increase in β_s always diminishes the primary product price m_S .

Proof of Corollary 2 We first check how the monopoly firm's profit changes when β_s increases. Calculating the derivation of the firm G 's profit with respect to β_s , we derive:

$$\frac{d\pi_G}{d\beta_s} = \frac{(\beta_s + 2t)(\beta_s^2 + 4\beta_s t - 9t^2 + 6tv^s)}{144t^2}.$$

Since the denominator is always positive, the sign of $\frac{d\pi_G}{d\beta_s}$ is determined by the numerator $(\beta_s + 2t)(\beta_s^2 + 4\beta_s t - 9t^2 + 6tv^s)$. Under Assumption 1, the equation $\beta_s^2 + 4\beta_s t - 9t^2 + 6tv^s$ is always positive.

Therefore, $\frac{d\pi_G}{d\beta_s}$ is positive if $\beta_s > -2t$ and negative otherwise.

As for the profit of fringe firms, its derivative equals

$$\frac{d\pi_C}{d\beta_s} = \frac{(\beta_s - 3t)(2\beta_s^2 + 3\beta_s t + 3t(-5t + 2v^s))}{216Nt^2}.$$

Since $|\beta_s| < 3t$ under Assumption 1, equation $(\beta_s - 3t)$ is always negative. Hence, the sign of $d\pi_C/d\beta_s$ is determined by the convex function $H_C = 2\beta_s^2 + 3\beta_s t + 3t(-5t + 2v^s)$ in β_s . We calculate the function's minimum value at $\beta_s = -3t/4$ and derive positive value under Assumption 1. Consequently, the profit of competing firms always diminishes following self-preferencing.

Proof of Corollary 3 Calculating the derivative of n_S with respect to β_s , we obtain:

$$\frac{dn_S}{d\beta_s} = \frac{\beta_s + 2t}{12t}.$$

As the denominator is always positive, the sign of the derivative is determined by its numerator. Correspondingly, $dn_S/d\beta_s$ is negative if $\beta_s < -2t$, and it's positive otherwise.

As for buyer participation, its derivative with respect to β_s equals:

$$\frac{dn_B}{d\beta_s} = \frac{\beta_b}{6t}.$$

Therefore, the sign of $dn_B/d\beta_s$ is positive when $\beta_b > 0$, and is the opposite otherwise.

A.5 MSM Estimation

To provide counterfactual exercises regarding alternative regulatory interventions, we estimate key model parameters using the Method of Simulated Moments (MSM). The estimation is conducted using data from the top 100 most popular news and media websites in France, with a focus on two primary outcomes: the count of advertisers posting ads on each website, and the advertising revenue earned by each publisher. These outcomes are observed both before and after the FCA regulation, allowing us to capture shifts in market outcomes due to regulatory changes.

As the traffic to news websites varies and is relatively concentrated, we introduce heterogeneity across websites to generate realistic simulated data. Specifically, we assume that each publisher's value-relevant parameter for adtech, v^s , follows a Pareto distribution characterized by parameters μ_{vs} and σ_{vs} . Similarly, each advertiser's parameter regarding their standalone value from Google's adtech, v_G^b , is drawn from a Pareto distribution governed by μ_{vGb} and σ_{vGb} . For both distributions, the scale parameter μ determines the location of the distribution, while the shape parameter σ governs the tail behavior, affecting the degree of variation and the frequency of extreme values. These distributions reflect the cross-sectional variation observed in the market and enable the model to produce diverse outcomes for advertisers and publishers across websites. The transportation cost parameter t , representing horizontal differentiation, is normalized at 1 throughout the estimation procedure.

The parameter vector we aim to estimate includes eight parameters.

$$\phi = [\beta_s^{pre}, \beta_s^{post}, \beta_b, k, \mu_{vs}, \sigma_{vs}, \mu_{vGb}, \sigma_{vGb}].$$

Among them, the distributional parameters ($\mu_{vs}, \sigma_{vs}, \mu_{vGb}, \sigma_{vGb}$) determine the spread and central tendency of advertiser and publisher heterogeneity. The structural parameters include β_s^{pre} and β_s^{post} , which capture the vertical quality gap between the 1P service and competing 3P alternatives for publishers before and after the regulation. We also estimate β_b , representing advertisers' relative value of the 1P service over rivals, and a scale factor k that converts normalized user shares in the model into actual advertiser counts observed in the data.

The estimation strategy relies on matching eight empirical moments: the mean and variance of (i) advertiser counts and (ii) publisher revenues, each observed *before* and *after* the FCA regulation. Formally, the empirical moment vector is defined as:

$$m_{emp} = [\mathbb{E}[n_B^{pre}], \text{Var}(n_B^{pre}), \mathbb{E}[R^{pre}], \text{Var}(R^{pre}), \mathbb{E}[n_B^{post}], \text{Var}(n_B^{post}), \mathbb{E}[R^{post}], \text{Var}(R^{post})],$$

where n_B is the advertiser count, R is the publisher revenue, and the superscription denotes the corresponding period before or after the FCA regulation.

The simulated counterpart $m_{sim}(\phi)$ is constructed analogously from model-generated data. Since Google charges fees from both publishers and advertisers, our estimation is based on the previous model that incorporates both buyer and seller fees. Specifically, the model-generated advertiser count is directly obtained from Equation 19:

$$n_B = k \cdot \frac{\beta_b \beta_s - 3\beta_b t + 6tv_G^b}{12t}.$$

As for the advertising revenue of publishers, it is mapped to the expected profit of sellers before paying corresponding fees for primary products or ancillary services. Therefore, it can be calculated as:

$$\begin{aligned} R &= \sum_{k=1}^N \left(\int_0^{\bar{\theta}} (v_G^s - tx) dx + \int_{\bar{\theta}}^1 (v_G^s - \beta_s - t(1-x)) dx \right) \frac{1}{N}, \\ &= \frac{5\beta_s^2 - 9t^2 + 18tv^s}{36t}. \end{aligned} \quad (16)$$

These equations are evaluated both before and after the policy intervention, using the corresponding parameter values in ϕ . For each parameter guess, we simulate the model 1,000 times. In each simulation, we first draw 100 pairs of random values (v^s and v_G^b) from the Pareto distribution for 100 websites, then compute advertiser participation and publisher revenue based on the model's equilibrium equations. We next calculate the mean and variance of advertiser count and publisher revenue across the 100 websites. Averaging across the 1,000 simulations yields a vector of simulated moment vector m_{sim} . Lastly, we compare to the empirical moment vector m_{emp} and the simulated moment m_{sim} , minimizing the distance between these two vectors in a weighted quadratic form. Specifically, the objective function Obj is defined as a quadratic form:

$$Obj = (m_{sim} - m_{emp})' W (m_{sim} - m_{emp}),$$

where W is a diagonal weighting matrix constructed as the inverse square of each empirical moment. This weighting matrix ensures that all moment discrepancies are scaled proportionally to their empirical magnitudes.

The estimation is carried out in two steps. In each step, we impose constraints to ensure economically meaningful outcomes: (i) the parameters must satisfy the assumptions of our model;

(ii) the simulated means of advertiser count, publisher revenue, and the estimated self-preferencing measurement β_s must decrease following regulation. In the first step, we use a diagonal weighting matrix. The constrained optimization is performed using the Sequential Least Squares Programming (SLSQP). After obtaining initial parameter estimates from the first step, we proceed to the second step by updating the weighting matrix. This is done by simulating the model using the step-one parameter estimates, computing the covariance matrix of the simulated moments, and taking its inverse to form a new weighting matrix. We then re-run the optimization using the updated weights.

To ensure robustness and avoid local minima, we explore 100 random initializations for the parameter vector and select the parameter set that minimizes the objective function as the optimal output. Our final parameter estimates yield an objective value lower than 0.001 and the estimated parameters are as follows:

$$\hat{\phi} = [1.39, 0.11, 3.59, 271.38, 4.24, 2578.07, 2.94, 909.44].$$

Based on the estimation, we then use the calibrated model to simulate counterfactual policy scenarios and compute the resulting changes in total welfare.³⁴ Precisely, we consider three main scenarios: (i) the actual behavioral remedy implemented by the FCA, (ii) an enhanced behavioral remedy that improves 3P service value for both sides of the market, and (iii) a structural remedy involving a divestment of Google's ancillary service (AdX). In each case, we simulate the market using the estimated parameters and compute the ratio of total surplus change after the policy intervention. Precisely, our welfare metric W focuses on the surplus sum of buyers and sellers:

$$w = Surplus_B^* + Surplus_S^* = \frac{1 + (n_B^*)^2}{2} + \frac{1 + (n_S^*)^2}{2},$$

where n_B^* and n_S^* follows the same format as Equation 19 and Equation 20 in the first two scenarios. In the third scenario, the profit of the dominant firm changes, and the equilibrium participation becomes:

$$\begin{aligned} n_B^* &= \frac{\beta_b \beta_s - 3\beta_b t + 6tv_G^b}{12t}, \\ n_S^* &= \frac{\beta_s^2 - 45t^2 + 18tv^s}{72t}. \end{aligned}$$

Let w_{pre} and w_{post} denote the total surplus before and after a policy intervention. The welfare change

³⁴All the estimated parameters satisfy our model assumption.

ratio in Figure 4 is computed as:

$$\Delta W = \frac{w_{\text{post}} - w_{\text{pre}}}{w_{\text{pre}}} \times 100.$$

In all three scenarios, the vertical differentiation between the 1P and 3P services changes from $\beta_s^{\text{pre}} = 1.39$ to $\beta_s^{\text{post}} = 0.11$ after the regulation. Since the latter two scenarios involve a reduction in vertical differentiation between the 1P and 3P services for advertisers, we analyze different cases in which this differentiation is diminished by varying degrees.

A.6 Alternative Remedies in Theoretical Model

We now map the two alternative remedies to our theoretical model. We begin by analyzing the structural divestment. As illustrated in the main text, we decompose the overall changes in market outcomes following the divestment as follows:

$$\begin{aligned}\Delta EQ &= EQ'(\beta'_s, \beta'_b) - EQ(\beta_s, \beta_b), \\ &= EQ'(\beta'_s, \beta'_b) - EQ(\beta'_s, \beta'_b) + EQ(\beta'_s, \beta'_b) - EQ(\beta_s, \beta_b), \\ &\approx \underbrace{EQ'(\beta'_s, \beta'_b) - EQ(\beta'_s, \beta'_b)}_{\text{induced by market structure adjustment}} + \underbrace{\frac{\partial EQ(\beta_s, \beta_b)}{\partial \beta_b}(\beta'_b - \beta_b)}_{\text{induced by } \beta_b \text{ adjustment}} + \underbrace{\frac{\partial EQ(\beta_s, \beta_b)}{\partial \beta_s}(\beta'_s - \beta_s)}_{\text{induced by } \beta_s \text{ adjustment}}.\end{aligned}$$

To study the first effect, we analyze the scenario where the ownership of the 1P ancillary product is switched from firm G to a different company, denoted as firm B . After this divestment, firm G collects profit only from the primary market. Maximizing its profit, we derive the market equilibrium. To investigate the welfare implications, we compare user participation before and after the divestment in a market with network effects. On the seller side, the change in market participation is given by:

$$\Delta n_S^I = n_S^{d*} - n_S^* = -\frac{(\beta_s + 3t)^2}{36t(1 - e_b e_s)} < 0,$$

where n_S^{d*} denotes the seller participation after the divestment, and Δn_S^I measures the change in seller participation induced by market structure modification. Likewise, on the buyer side, we find that:

$$\Delta n_B^I = n_B^{d*} - n_B^* = -\frac{e_b(\beta_s + 3t)^2}{36t(1 - e_b e_s)} < 0,$$

where n_B^{d*} is buyers' participation after the divestment, and Δn_B^I measures the change in buyer participation induced by market structure modification.

As for the competing firms in the adjacent market, the change in their profit following divestment is given by:

$$\pi_C^{d*} - \pi_C^* = -\frac{(\beta_s^2 - 9t^2)^2}{648Nt^2(1 - e_b e_s)} < 0,$$

where π_C^{d*} denotes the profit of competing firms after the divestment.

The results suggest that the first impact alone, which is induced by the market structure modification following divestment, always diminishes user surplus and their participation in the market. This is

because divestment decreases the profit each user brings to the gatekeeper firm. Consequently, firm G 's incentive to expand market size diminishes, leading to decreased aggregate demand for market participation. Moreover, competing firms tend to experience lower profits after divestment. In the long run, the negative impact on competing firms may speed up their exit, ultimately reducing overall competition in the ancillary product market.

To investigate the second effect, we examine the marginal change in market participation when the value of 3P products is improved for buyers. Plugging in Equations 17-18, we obtain:

$$\begin{aligned}\Delta n_B^{II} &= \frac{\partial n_B^*}{\partial \beta_b}(\beta'_b - \beta_b) = \frac{(2 - e_b e_s)(\beta_s - 3t)}{12(1 - e_b e_s)t}(\beta'_b - \beta_b) > 0, \\ \Delta n_S^{II} &= \frac{\partial n_S^*}{\partial \beta_b}(\beta'_b - \beta_b) = \frac{e_s(\beta_s - 3t)}{12(1 - e_b e_s)t}(\beta'_b - \beta_b) > 0.\end{aligned}$$

We find that the second effect, which is induced by a potential improvement in buyers' value for 3P ancillary products following divestment, always positively affects user surplus on both sides of the market.

In summary, compared to a behavioral remedy whose impacts are largely captured in the third component of Equation 14, the divestment regulation generates two additional and conflicting effects on social welfare. As a result, there is no clear answer regarding which regulatory approach is superior. However, our analysis underscores the importance for policymakers to understand how the value of 3P ancillary products is likely to change after the divestment. If the divestment intervention only increases the value of 3P products for sellers without affecting buyers—i.e., when the second effect is absent—such a structural remedy could reduce social welfare to a greater extent than a behavioral regulation. However, if the divestment intervention increases the value of 3P products for both buyers and sellers and generates a sufficiently large second effect, the divestment remedy may improve social welfare more effectively than the behavioral regulation.

We next study the behavioral remedy of improving 3P ancillary services on both sides. To illustrate, we investigate how n_B^* and n_S^* (Equation 19 and Equation 20) evolve when 3P ancillary services improve for buyers (advertisers) and sellers (publishers):

$$\begin{aligned}\Delta n_B &= \frac{\partial n_B^*}{\partial \beta_b} \Delta \beta_b + \frac{\partial n_B^*}{\partial \beta_s} \Delta \beta_s = \frac{(\beta_s - 3t)}{12t} \Delta \beta_b + \frac{\beta_b}{12t} \Delta \beta_s, \\ \Delta n_S &= \frac{\partial n_S^*}{\partial \beta_b} \Delta \beta_b + \frac{\partial n_S^*}{\partial \beta_s} \Delta \beta_s = \frac{(\beta_s + 2t)}{12t} \Delta \beta_s,\end{aligned}$$

where Δn_B and Δn_S are the approximate changes in user participation, and $\Delta\beta_s < 0$ and $\Delta\beta_b < 0$ measure the improvement of 3P ancillary services for publishers and advertisers.

On the publisher side, improving 3P ancillary services for advertisers has no impact on publishers, while improving 3P ancillary services for publishers diminishes publisher participation and their surplus when 1P ancillary services are of higher value than 3P ancillary services, i.e., $\beta_s > 0$. Hence, in the context of the Google case, the overall impact on publisher surplus is typically negative following regulation.

On the advertiser side, there are two conflicting effects. On the one hand, when advertisers prefer 1P ancillary services over 3P ancillary services, i.e., $\beta_b > 0$, improving 3P ancillary services for publishers reduces advertiser participation and their surplus. On the other hand, improving 3P ancillary services for advertisers improves advertiser participation and their surplus under Assumption 2. Hence, compared to the FCA remedy that only improves 3P ancillary services for publishers, the improvement in 3P ancillary services for advertisers introduces one additional positive impact on advertiser surplus.

In summary, the overall impact on total welfare can be either negative or positive, depending on the magnitude of these potentially conflicting effects. The results are highly consistent with our counterfactual simulations in the main text.

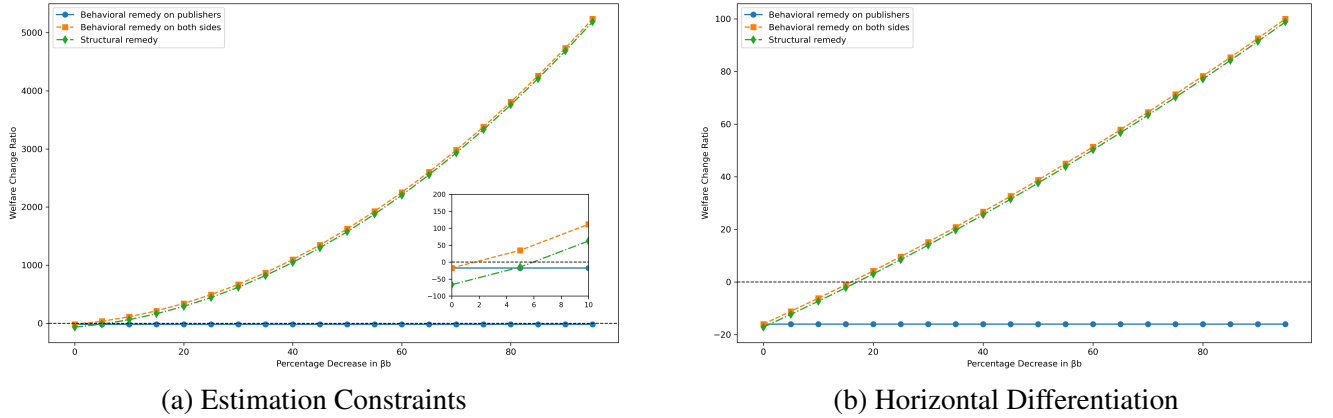
A.7 Robustness Checks for Counterfactual Analysis

To investigate the robustness of our findings in Figure 4, we provide the following robustness checks.

In Figure A.5a, we relax the estimation constraints which require that the simulated advertiser count, publisher revenue, and the estimated self-preferencing measurement β_s decrease following regulation. Without such a constraint, the simulated publisher revenue based on the estimation results increases, albeit by small magnitudes, after the FCA regulation. In this case, the qualitative changes in the simulated and empirical moments following regulation are misaligned. The changes in seller surplus and publisher revenue move in opposite directions, so the magnitude of the total welfare change as shown in the zoomed-in figure of Figure A.5a is much smaller than in the main text with the estimation constraints.

In Figure A.5b, we explore the welfare change ratio when we adjust the assumption of horizontal differentiation from $t = 1$ to $t = 2$, while all other settings remain the same in the previous estimation. We find that an increase in horizontal differentiation may reduce the welfare changes after regulation. With larger horizontal differentiation, there is less competition among firms in ancillary services, and alleviating self-preferencing may generate smaller welfare changes in magnitude. In both cases, although the qualitative estimation of welfare change varies when we change the horizontal differentiation assumption or relax the estimation constraints, the qualitative conclusions regarding the welfare efficiency across the three remedies remain robust.

Figure A.5: Welfare Changes



B Online Appendix

B.1 Markets with Network Effects

We now solve for the subgame perfect equilibrium for a market with network effects. To avoid trivial market equilibria, we make the following assumptions in this section:

Assumption 2. *In a two-sided market with network effects, we assume:*

- I) [Profit Concavity] the network effects satisfy $e_s > 0$, $e_b > 0$, and $e_b e_s < 1$ to guarantee profit concavity;*
- II) [Positive Demand] the ancillary service values satisfy $|\beta_s| < 3t$ so that both 1P service and 3P service derive a positive demand at equilibrium; and*
- III) [Competitive Market and Market Participation] the surplus v^s and v_G^b satisfy the conditions such that there exists competition between the 1P service and 3P service, and user entry does not hit corner solution.*

The introduction of network effects simultaneously increases the value of all users. Hence, it does not affect the competition or transaction stage in the adjacent market. The demand and firms' pricing strategy over ancillary services stay the same as in the benchmark. However, buyers and sellers' market participation change. Precisely, a seller's expected surplus from joining the market, denoted as $E[U_s]$, equals:

$$E[U_s] = \sum_{k=1}^N \left(\int_0^{\bar{\theta}(k)} (v_G^s + e_s n_B - p_G - tx) dx + \int_{\bar{\theta}(k)}^1 (v_G^s + e_s n_B - \beta_s - p_k - t(1-x)) dx \right) \frac{1}{N} - m_S.$$

As for buyers, their expected utility depends on the distribution of sellers among ancillary services and can be written as:

$$E[U_b] = \sum_{k=1}^N \left(\int_0^{\bar{\theta}(k)} (v_G^b + e_b n_S) dx + \int_{\bar{\theta}(k)}^1 (v_G^b - \beta_b + e_b n_S) dx \right) \frac{1}{N}.$$

Users in both groups will join the market if they expect to derive a higher surplus than their outside options. Hence, the demands for market participation in both groups can be written as $n_S = E[U_s]$ and $n_B = E[U_b]$.

Given the ancillary service prices (expression 2-3) and demands (expression 4-5) at equilibrium, the explicit demands for market participation can be expressed as:

$$\begin{aligned} n_B &= \frac{6\beta_b(\beta_s - 3t) + 36tv_G^b + e_b\beta_s^2 - 9te_b(4m_S + 5t - 2v^s)}{36t(1 - e_be_s)}, \\ n_S &= \frac{\beta_s^2 + 6e_s(\beta_b\beta_s - 3t\beta_b + 6tv_G^b) - 9t(4m_S + 5t - 2v^s)}{36t(1 - e_be_s)}. \end{aligned}$$

Maximizing firm G 's profit by solving the first-order condition, we derive the profit-maximizing primary product price as:

$$m_S^* = \frac{6e_s\beta_b\beta_s - \beta_s^2 - 18e_s\beta_bt - 12\beta_st - 63t^2 + 36e_stv_G^b + 18tv^s}{72t}.$$

Accordingly, the equilibrium demands for market participation are given by:

$$n_B^* = \frac{2(2 - e_be_s)\beta_b(\beta_s - 3t) + 24tv_G^b + e_b(\beta_s^2 + 4\beta_st - 9t^2 - 12te_stv_G^b + 6tv^s)}{24t(1 - e_be_s)}, \quad (17)$$

$$n_S^* = \frac{2e_s\beta_b\beta_s + \beta_s^2 - 6e_s\beta_bt + 4\beta_st - 9t^2 + 12e_stv_G^b + 6tv^s}{24t(1 - e_be_s)}, \quad (18)$$

and the resulting demands for ancillary services are $Q_G^* = (\frac{1}{2} + \frac{\beta_s}{6t})n_S^*$ and $Q_C^* = (\frac{1}{2} - \frac{\beta_s}{6t})\frac{n_S^*}{N}$.

We lastly calculate the equilibrium profits of firms and derive:

$$\begin{aligned} \pi_G^* &= \frac{(2e_s\beta_b\beta_s + \beta_s^2 - 6e_s\beta_bt + 4\beta_st - 9t^2 + 12e_stv_G^b + 6tv^s)^2}{576t^2(1 - e_be_s)}, \\ \pi_C^* &= \frac{(3t - \beta_s)^2(2e_s\beta_b\beta_s + \beta_s^2 - 6e_s\beta_bt + 4\beta_st - 9t^2 + 12e_stv_G^b + 6tv^s)}{432Nt^2(1 - e_be_s)} - L. \end{aligned}$$

To investigate the impact of self-preferencing on market competition, we calculate the derivative of n_S with respect to β_s and obtain:

$$\frac{dn_S}{d\beta_s} = \frac{e_s\beta_b + \beta_s + 2t}{12t(1 - e_be_s)}.$$

Under Assumption 2, the value of $\frac{dn_S}{d\beta_s}$ is increasing in β_b . Specifically, $\frac{dn_S}{d\beta_s}$ is negative if $\beta_s <$

$-2t - e_s\beta_b$, and it's positive if $\beta_s > -2t - e_s\beta_b$.

As for the buyer participation, its derivative with respect to β_s equals:

$$\frac{dn_B}{d\beta_s} = \frac{2\beta_b - e_b e_s \beta_b + e_b \beta_s + 2e_b t}{12t(1 - e_b e_s)}.$$

Under Assumption 2, it is easy to show that the value of $\frac{dn_B}{d\beta_s}$ is increasing in β_b . Specifically, $\frac{dn_B}{d\beta_s}$ is positive when $\beta_b > -\frac{e_b(2t+\beta_s)}{2-e_b e_s}$ and negative otherwise.

We next check how firm G 's profit changes when β_s increases. Calculating the derivation of firm G 's profit with respect to β_s , we derive:

$$\frac{d\pi_G}{d\beta_s} = \frac{(e_s\beta_b + \beta_s + 2t)(2e_s\beta_b\beta_s + \beta_s^2 - 6e_s\beta_b t + 4\beta_s t - 9t^2 + 12e_s t v_G^b + 6t v^s)}{144t^2(1 - e_b e_s)}.$$

Since the denominator is always positive, the sign of $\frac{d\pi_G}{d\beta_s}$ is determined by the numerator $(e_s\beta_b + \beta_s + 2t)(2e_s\beta_b\beta_s + \beta_s^2 - 6e_s\beta_b t + 4\beta_s t - 9t^2 + 12e_s t v_G^b + 6t v^s)$. Under Assumption 2, the equation $2e_s\beta_b\beta_s + \beta_s^2 - 6e_s\beta_b t + 4\beta_s t - 9t^2 + 12e_s t v_G^b + 6t v^s$ is always positive. Therefore, $\frac{d\pi_G}{d\beta_s}$ is positive if $\beta_s > -2t - e_s\beta_b$ and negative otherwise.

B.2 Market with Fees on Both Sides

We now solve for the subgame perfect equilibrium when the gatekeeper charges also buyers. The introduction of buyer fee does not affect the transaction stage in the adjacent market, thus the demand and firms' pricing strategy over ancillary services stay the same as in the benchmark. However, the market participation of buyers and sellers change.

When buyers also need to pay the gatekeeper m_B for joining the market, the surplus of users become:

$$\begin{aligned} E[U_s] &= -m_S + \sum_{k=1}^N \left(\int_0^{\bar{\theta}(k)} (v_G^s - p_G - tx) dx + \int_{\bar{\theta}(k)}^1 (v_G^s - \beta_s - p_k - t(1-x)) dx \right) \frac{1}{N}, \\ E[U_b] &= -m_B + \sum_{k=1}^N \left(\int_0^{\bar{\theta}(k)} v_G^b dx + \int_{\bar{\theta}(k)}^1 (v_G^b - \beta_b) dx \right) \frac{1}{N}. \end{aligned}$$

Correspondingly, the explicit demands can be simplified as:

$$\begin{aligned} n_B &= v_G^b - m_B + \frac{\beta_b(\beta_s - 3t)}{6t}, \\ n_S &= \frac{\beta_s^2 - 9t(4m_S + 5t - 2v^s)}{36t}. \end{aligned}$$

We next examine firm G 's optimal pricing strategy. In this context, it charges both buyers and sellers. Hence, its profit (π_G) is expressed as:

$$\pi_G = m_B n_B + m_S n_S + \sum_{k=1}^N p_G \frac{(\beta_s - p_G + p_k + t) n_S}{2Nt}.$$

Maximizing its profit, the gatekeeper's equilibrium price in the primary market can be derived as follows.

$$\begin{aligned} m_B^* &= \frac{\beta_b \beta_s - 3\beta_b t + 6t v_G^b}{12t}, \\ m_S^* &= \frac{18t v^s - \beta_s^2 - 12\beta_s t - 63t^2}{72t}. \end{aligned}$$

To investigate the welfare implication of self-preferencing, we calculate the equilibrium demands

for market participation and derive:

$$n_B^* = \frac{\beta_b \beta_s - 3\beta_b t + 6tv_G^b}{12t}, \quad (19)$$

$$n_S^* = \frac{\beta_s^2 + 4\beta_s t - 9t^2 + 6tv^s}{24t}. \quad (20)$$

Analyzing how n_B^* and n_S^* evolve when firm G practices self-preferencing, we find qualitatively similar findings to Corollary 3.

$$\frac{\partial n_B^*}{\partial \beta_s} = \frac{\beta_b}{12t}, \quad (21)$$

$$\frac{\partial n_S^*}{\partial \beta_s} = \frac{\beta_s + 2t}{12t}. \quad (22)$$

As for the impact of self-preferencing on firm G 's profit, the corresponding derivative can be written as:

$$\frac{\partial \pi_G^*}{\partial \beta_s} = \underbrace{\frac{2\beta_b(\beta_b \beta_s - 3\beta_b t + 6tv_G^b)}{144t^2}}_{\text{First Component}} + \underbrace{\frac{(\beta_s + 2t)(\beta_s^2 + 4\beta_s t - 9t^2 + 6tv^s)}{144t^2}}_{\text{Second Component}}.$$

Following the same steps as in the benchmark, we can easily show that the second component alone derives a similar conclusion as in the benchmark. Precisely, the value of the second component is positive when $\beta_s > -2t$ and is negative otherwise. As for the first component, the value of $(\beta_b \beta_s - 3\beta_b t + 6tv_G^b)$ is always positive as long as there are positive number of buyers entering the market. Thus, if $\beta_b > 0$, the first component is always positive. Accordingly, seller's preference cutoff $\tilde{\beta}_s$ above which the gatekeeper firm chooses to practice self-preferencing must decrease and be lower than $-2t$. In contrast, if $\beta_b < 0$, the first component is always negative. Accordingly, the cutoff above which the gatekeeper firm chooses to practice self-preferencing must increase and be higher than $-2t$.

B.3 Long-Run Impact

We now analyze the context where the gatekeeper monopolizes the ancillary service market. To simplify, we focus on the case where $\beta_b = 0$, namely, buyers perceive all sellers and ancillary services as identical. In such a market, the demand for ancillary services can be expressed as follows:

$$Q_G = \begin{cases} \frac{(v_G^s - p_G - m_S)n_S}{t} & \text{if } p_G + m_S > v_G^s - t, \\ n_S & \text{if } p_G + m_S \leq v_G^s - t. \end{cases}$$

For each seller entering the market, they must pay the gatekeeper for both the primary product and ancillary service. In other words, the primary product and ancillary service are now bundled, with a price of $p_G + m_S$. When such a bundle price is high, the market operates as a local monopoly with a partially covered market. When the price is low, the market becomes fully covered.

Following the same steps as in the benchmark, we then calculate sellers' expected surplus from entering the market and obtain the following seller participation:

$$n_S = \begin{cases} \frac{(v_G^s - p_G - m_S)^2}{2t} & \text{if } p_G + m_S > v_G^s - t, \\ \frac{t}{2} & \text{if } p_G + m_S \leq v_G^s - t. \end{cases}$$

We begin by investigating the partially covered market. In this scenario, the gatekeeper's profit is given by:

$$\pi_G = (p_G + m_S)Q_G = \frac{(p_G + m_S)(v_G^s - p_G - m_S)^3}{2t^2}.$$

Solving the first-order condition, we derive the optimal pricing strategy: $p_G + m_S = \frac{v_G^s}{4}$. Accordingly, the equilibrium demands for market participation are:

$$n_S^* = \frac{9(v_G^s)^2}{32t}, \quad n_B^* = v_G^b.$$

When sellers have a high value from entering the market, such that $v_G^s \geq \frac{4t}{3}$, the market becomes fully covered. In this scenario, the gatekeeper's profit becomes:

$$\pi_G = \frac{(p_G + m_S)t}{2}.$$

It is optimal for the gatekeeper to continue increasing its price until the seller, whose preferred service would be 3P offerings, derives zero surplus. In other words, the bundle price of the primary product and ancillary service equals $p_G + m_S = v_G^s - t$, and the resulting profit of the gatekeeper is:

$$\pi_G^* = \frac{t(v_G^s - t)}{2}.$$

The corresponding equilibrium demands for market participation are:

$$n_S^* = \frac{t}{2}, \quad n_B^* = v_G^b.$$

To summarize, once the gatekeeper firm monopolizes the ancillary service market, buyer participation becomes $n_B^* = v_G^b$ and seller participation can be written as:

$$n_S^* = \begin{cases} \frac{9(v_G^s)^2}{32t} & \text{if } v_G^s < \frac{4t}{3}, \\ \frac{t}{2} & \text{if } v_G^s \geq \frac{4t}{3}. \end{cases}$$

Given the market equilibrium, we first compare the monopoly case with the benchmark when $v_G^s < \frac{4t}{3}$. In this scenario, the difference in seller participation can be calculated as:

$$\Delta n_S = n_S^* - n_S^m = \frac{4\beta_s^2 - 8\beta_s t - 36t^2 + 48tv_G^s - 27(v_G^s)^2}{96t},$$

where n_S^* is the seller participation in the benchmark model and n_S^m is in the monopoly market.

Since the seller participation serves as an indicator of their surplus, we are interested in the value of Δn_S . Specifically, the equation Δn_S first increases in v_G^s and then decreases within the interval $\beta_s < v_G^s < \frac{4t}{3}$. To investigate, we first calculate Δn_S at the lower bound, where $v_G^s = \beta_s$. The condition $v_G^s > \beta_s$ ensures a positive 3P ancillary service value. We obtain the expression $\Delta n_S = -23\beta_s^2 + 40t\beta_s - 36t^2$, which is always negative under $t > 0$. Next, we examine Δn_S at the upper bound, where $v_G^s = \frac{4t}{3}$. We find that $\Delta n_S = 4(\beta_s^2 - 2\beta_s t - 5t^2)$ remains positive, except when $\beta_s > (1 - \sqrt{6})t$. Hence, unless the 3P ancillary service quality falls below a certain threshold ($v_G^s - \beta_s < v_G^s - (1 - \sqrt{6})t$), there exists a threshold $\tilde{v}_G^s$ such that $\Delta n_S > 0$ when $\frac{4t}{3} > v_G^s > \tilde{v}_G^s$. This indicates a reduction in seller surplus when the market shifts from a competitive (benchmark) structure to a local monopoly.

We then compare the monopoly case with the benchmark when $v_G^s > \frac{4t}{3}$. As the seller participation in the benchmark increases with v_G^s , while it is unaffected by v_G^s in the monopoly market, we can

easily show that both the seller participation and their surplus diminish once the market switches from competitive to monopoly, provided that their value for market participation (v_G^s) is not too low.

In summary, from the seller's perspective, switching from a competitive market to a monopoly in the long run reduces their surplus, as long as their value for entering the market (v_G^s) is not too low. For buyers, when $\beta_b = 0$ and no network effects are present, there is no change in surplus. However, when positive network effects are considered, the reduction in seller participation negatively affects buyers through these network effects, ultimately diminishing buyer surplus.

B.4 Horizontal Differentiation

In the main text, we assume that the horizontal differentiation between the 1P and 3P services is fixed and unaffected by firms' entry into the adjacent market. In this section, we relax this assumption and explore whether the main conclusions still hold.

To incorporate the impact of firms' entry on the horizontal differentiation between ancillary services, we denote the length of the Hotelling line as $b(N)$, a function of the number of competing firms in the adjacent market. Accordingly, the density of seller mass becomes $1/b(N)$ to normalize the user size. All other setups remain the same as the benchmark. Within this model setup, we then solve for the subgame perfect equilibrium.

If firm G and firm $k \in \{1, 2, \dots, N\}$ are in a seller's consideration set, the seller is indifferent between the two firms when his location $\bar{\theta}(k)$ satisfies $U_S(\bar{\theta}(k), G) = U_S(\bar{\theta}(k), k)$. Solving this equation, we derive the indifferent seller's location as follows:

$$\bar{\theta}(k) = \frac{\beta_s + p_k - p_G + bt}{2t}.$$

Correspondingly, the demands for the two ancillary services can be expressed as:

$$\begin{aligned} Q_G &= \sum_{k=1}^N \bar{\theta}(k) \frac{1}{b} \frac{n_S}{N} = \sum_{k=1}^N \frac{(\beta_s + p_k - p_G + bt)}{2t} \frac{n_S}{bN}, \\ Q_k &= (1 - \bar{\theta}(k)) \frac{1}{b} \frac{n_S}{N} = (b - \frac{\beta_s + p_k - p_G + bt}{2t}) \frac{n_S}{bN}. \end{aligned}$$

Given the prices, buyers and sellers make entry decisions and their demand for market participation can be solved as:

$$\begin{aligned} n_S &= -m_S + \sum_{k=1}^N \left(\int_0^{\bar{\theta}(k)} (v_G^s - p_G - tx) dx + \int_{\bar{\theta}(k)}^b (v_G^s - \beta_s - p_k - t(1-x)) dx \right) \frac{1}{bN}, \\ n_B &= \sum_{k=1}^N \left(\int_0^{\bar{\theta}(k)} v_G^b dx + \int_{\bar{\theta}(k)}^b (v_G^b - \beta_b) dx \right) \frac{1}{bN}. \end{aligned}$$

Each firm maximizes its profit simultaneously and the equilibrium prices are given by:

$$\begin{aligned} p_G &= bt + \frac{\beta_s}{3}, \\ p_C &= bt - \frac{\beta_s}{3}, \end{aligned}$$

and the resulting ancillary service demands are:

$$\begin{aligned} Q_G &= \left(\frac{b}{2} + \frac{\beta_s}{6t}\right)n_S, \\ Q_C &= \left(\frac{b}{2} - \frac{\beta_s}{6t}\right)\frac{n_S}{N}. \end{aligned}$$

Lastly, we proceed to analyze firm G 's pricing strategy over the primary product. Plugging in the equilibrium demands and prices, the explicit demands for market participation can be expressed as:

$$\begin{aligned} n_B &= \frac{\beta_b(\beta_s - 3bt) + 6btv_G^b}{6bt}, \\ n_S &= \frac{\beta_s^2 - 36btm_S - 45b^2t^2 + 18btv^s}{36bt}. \end{aligned}$$

Maximizing firm G 's profit by solving the first-order condition, we derive the profit-maximizing primary product price as:

$$m_S^* = \frac{-\beta_s^2 - 12\beta_sbt - 63b^2t^2 + 18btv^s}{72bt}.$$

Given the equilibrium pricing strategies, the demands for market participation are given by:

$$\begin{aligned} n_B^* &= \frac{\beta_b\beta_s + 6tbv_G^b - 3\beta_bbt}{6bt}, \\ n_S^* &= \frac{\beta_s^2 + 4\beta_sbt - 9b^2t^2 + 6btv^s}{24bt}, \end{aligned}$$

and the resulting equilibrium profits of firms are:

$$\begin{aligned} \pi_G^* &= \frac{(\beta_s^2 + 4\beta_sbt - 9b^2t^2 + 6btv^s)^2}{576bt^2}, \\ \pi_C^* &= \frac{(3bt - \beta_s)^2(\beta_s^2 + 4\beta_sbt - 9b^2t^2 + 6btv^s)}{432bNt^2} - L. \end{aligned}$$

Based on the market equilibrium, we first investigate firm G 's motivation for self-preferencing. Precisely, we find that the gatekeeper firm G derives a higher profit following self-preferencing if $\beta_s > -2bt$. Furthermore, analyzing market participation, we derive results qualitatively similar to the benchmark: sellers benefit from self-preferencing as long as it is profitable for firm G , i.e. when $\beta_s > -2bt$. However, buyers may be harmed by self-preferencing when they dislike the 1P service, i.e., $\beta_b < 0$, and benefit from it otherwise.

Note that we now allow b to be endogenously determined by firm entry N so the horizontal differentiation is also affected accordingly. However, since the cutoffs for both firm G and sellers to benefit from self-preferencing are equal, any change in b simply shifts the two cutoffs simultaneously without affecting their relationship. Therefore, allowing horizontal differentiation to vary with the number of competing firms in the ancillary service market does not alter the paper's main conclusion.

B.5 Consideration Set

In the main text, we assume that each seller considers two ancillary services: one is always the 1P service, and the other is a randomly selected 3P service. Thus there is no direct competition among 3P services. In this section, we relax this assumption and analyze the case where services in the consideration set are randomly chosen. In this context, fringe firms not only compete with the gatekeeper firm but also with each other in the adjacent market.

There are two types of indifferent sellers. If the gatekeeper firm G and a competitive fringe firm $k \in \{1, 2, \dots, N\}$ are in a seller's consideration set, the indifferent seller's location $\bar{\theta}(k)$ is the same as in the benchmark. If there are two competitive fringe firms $k, j \in \{1, 2, \dots, N\}$ in a seller's consideration set, the seller is indifferent between the two firms when he is located at:

$$\bar{\theta}^c(k, j) = \frac{p_k - p_j + t}{2t}.$$

Precisely, sellers will choose firm k 's ancillary service if $\theta > \bar{\theta}^c(k, j)$ and firm j 's ancillary service otherwise. Given that all firms, including the gatekeeper firm, all have an equal chance to appear in the consideration set, the demands for the 1P and 3P services can be expressed as:

$$\begin{aligned} Q_G &= \sum_{k=1}^N \frac{2n_s \bar{\theta}(k)}{N(N+1)}, \\ Q_k &= \frac{2n_s(1 - \bar{\theta}(k))}{N(N+1)} + \sum_{j \neq k \in \{1, 2, \dots, N\}} \frac{2n_s(1 - \bar{\theta}^c(k, j))}{N(N+1)} \end{aligned}$$

Given the product and service prices, buyers and sellers make entry decisions and their demand for market participation can be solved as:

$$\begin{aligned} n_S &= -m_S + \sum_{k=1}^N \left(\int_0^{\bar{\theta}(k)} (v_G^s - p_G - tx) dx + \int_{\bar{\theta}(k)}^1 (v_G^s - \beta_s - p_k - t(1-x)) dx \right) \frac{2}{N(N+1)} + \\ &\quad \sum_{k \neq j \in \{1, 2, \dots, N\}} \left(\int_0^{\bar{\theta}^c(k, j)} (v_G^s - \beta_s - p_j - t(1-x)) dx + \int_{\bar{\theta}^c(k, j)}^1 (v_G^s - \beta_s - p_k - t(1-x)) dx \right) \frac{1}{(N+1)N}, \\ n_B &= \sum_{k=1}^N \left(\int_0^{\bar{\theta}(k)} v_G^b dx + \int_{\bar{\theta}(k)}^1 (v_G^b - \beta_b) dx \right) \frac{2}{N(N+1)} + \frac{N(N-1)}{(N+1)N} (v_G^b - \beta_b). \end{aligned}$$

Each firm maximizes its profit and the ancillary service prices at the symmetric equilibrium are given

by:

$$\begin{aligned} p_G &= \frac{\beta_s N + t + 2Nt}{1 + 2N}, \\ p_C &= \frac{-\beta_s + t + 2Nt}{1 + 2N}. \end{aligned}$$

We then proceed to analyze firm G 's pricing strategy over the primary product. Plugging in the equilibrium demands and prices, the explicit demands for market participation can be expressed as:

$$\begin{aligned} n_B &= \frac{\beta_b N(\beta_s - (1 + 2N)t) + (1 + 3N + 2N^2)t v_G^b}{(1 + N)(1 + 2N)t}, \\ n_S &= \frac{2\beta_s^2 N^2 + 2\beta_s(1 + 3N - 4N^3)t - (1 + N)(1 + 2N)^2 t(4m_S + 5t - 2v^s)}{4(1 + N)(1 + 2N)^2 t}. \end{aligned}$$

Maximizing firm G 's profit by solving the first-order condition, we derive the profit-maximizing primary product price as:

$$m_S^* = \frac{(1 + 2N)^2 t(2(1 + N)v^s - (9 + 5N)t) - 2\beta_s^2 N^2 - 2\beta_s(-1 + N + 8N^2 + 4N^3)t}{8(1 + N)(1 + 2N)^2 t}.$$

Given the equilibrium pricing strategies, the demands for market participation are given by:

$$\begin{aligned} n_B^* &= \frac{\beta_b N(\beta_s - (1 + 2N)t) + (1 + 3N + 2N^2)t v_G^b}{(1 + N)(1 + 2N)t}, \\ n_S^* &= \frac{6\beta_s^2 N^2 + 2\beta_s(1 + 7N + 8N^2 - 4N^3)t - (1 + 2N)^2 t(t + 5Nt - 2(1 + N)v^s)}{8(1 + N)(1 + 2N)^2 t}, \end{aligned}$$

and the resulting equilibrium profits of the gatekeeper firm is:

$$\pi_G^* = \frac{(6\beta_s^2 N^2 + 2\beta_s(1 + 7N + 8N^2 - 4N^3)t - (1 + 2N)^2 t(t + 5Nt - 2(1 + N)v^s))^2}{64(1 + N)^2(1 + 2N)^4 t^2}.$$

Under the condition that there are positive market participation, the gatekeeper firm is motivated to practice self-preferencing when $\beta_s > -(1 + 7N + 8N^2 - 4N^3)t/6N^2$. As for the welfare implication, we derive qualitatively similar results to the benchmark: sellers benefit from self-preferencing as long as it is profitable for firm G , i.e. when $\beta_s > -(1 + 7N + 8N^2 - 4N^3)t/6N^2$. However, buyers may be harmed following self-preferencing when they dislike the 1P service, i.e., $\beta_b < 0$, and benefit from it

otherwise.

B.6 Outside Option

In the main text, we assume that both buyers' and sellers' outside options follow a uniform distribution on the interval $[0, 1]$. In this section, we relax this assumption. Specifically, we let $F_b(U)$ and $F_s(U)$ denote the cumulative distribution functions of buyers' and sellers' outside options, each is strictly increasing in U . Under this setup, the demands for market competition are given by $n_S = F_s(E[U_s])$ and $n_B = F_b(E[U_b])$.

The transaction stage in the ancillary service market remains the same as in the benchmark. In the primary product market, firm G 's equilibrium price is determined by the following first-order condition (FOC):

$$n_S + (m_S + p_G q_G) \frac{\partial n_S}{\partial m_S} = 0 \rightarrow n_S^* = - (m_s^* + p_G^* q_G^*) \frac{\partial F_s(E[U_s])}{\partial E[U_s]} \frac{\partial E[U_s]}{\partial m_S} = (m_s^* + p_G^* q_G^*) \frac{\partial F_s(E[U_s])}{\partial E[U_s]},$$

where $\frac{\partial E[U_s]}{\partial m_S} = -1$ if easily derived from Equation 1.

Accordingly, the derivative of seller participation with respect to β_s can be written as:

$$\frac{\partial n_S^*}{\partial \beta_s} = \frac{\partial (m_s^* + p_G^* q_G^*)}{\partial \beta_s} \frac{\partial F_s(E[U_s])}{\partial E[U_s]} + (m_s^* + p_G^* q_G^*) \frac{\partial F_s^2(E[U_s])}{\partial E[U_s] \partial \beta_s}.$$

Since the cumulative distribution function F_s must be increasing in the seller's expected surplus $E[U_s]$ and $(m_s^* + p_G^* q_G^*) > 0$, the effect of β_s on seller participation n_S^* depends on $\frac{\partial (m_s^* + p_G^* q_G^*)}{\partial \beta_s}$ and on $\frac{\partial F_s^2(E[U_s])}{\partial E[U_s] \partial \beta_s}$.

Only under specific conditions where $\frac{\partial (m_s^* + p_G^* q_G^*)}{\partial \beta_s}$ and $\frac{\partial F_s^2(E[U_s])}{\partial E[U_s] \partial \beta_s}$ have opposite signs, and the relative magnitudes satisfy:

$$\left| \frac{\partial F_s^2(E[U_s])}{\partial E[U_s] \partial \beta_s} \right| > \left| \frac{\frac{\partial (m_s^* + p_G^* q_G^*)}{\partial \beta_s}}{(m_s^* + p_G^* q_G^*)} \right|,$$

an increase in β_s moves n_S^* in the opposite direction to $(m_s^* + p_G^* q_G^*)$. In this extreme scenario, the gatekeeper's incentive and the sellers' surplus might become misaligned.

In all other cases, an increase in β_s will move n_S^* in the same direction as it moves $(m_s^* + p_G^* q_G^*)$. Recall that firm G 's profit is given by $\pi_G^* = (m_s^* + p_G^* q_G^*) n_S^*$. Since an increase in β_s affects both $(m_s^* + p_G^* q_G^*)$ and n_S^* in the same direction, we arrive at the same conclusion as in the benchmark: firm G 's incentive for self-preferencing aligns with seller participation and their surplus.

As for buyers, their aggregate surplus $E[U_b]$ is determined solely by sellers' distribution between the 1P and 3P services. Thus, the distribution of outside options affects only buyers' absolute level of market participation, not self-preferencing's impact on their surplus.

In summary, our key findings in Corollary 4 remain robust for most scenarios except one extreme case.

B.7 Alternative Self-Preferencing Definition

In the main text, we assume that self-preferencing does not affect the total value associated with the 1P and 3P services, i.e., $v_G^s + v_k^s = v^s$. In this section, we relax this assumption, allowing self-preferencing to affect not only the value distribution between the two ancillary services but also their total sum. Specifically, we assume $v_G^s + v_k^s = v^s - z\beta_s$, where $0 \leq z \leq 1$. Precisely, this alternative definition can be summarized as below:

Definition 2. *Self-preferencing refers to a practice by which the gatekeeper increases the relative value of its 1P ancillary service compared to competing 3P ancillary services, while holding the total ancillary value as $v_G^s + v_k^s = v^s - z\beta_s$. In our framework, self-preferencing corresponds to a marginal increase in the vertical differentiation parameter β_s , such that $v_G^s = (v^s + \beta_s - z\beta_s)/2$ and $v_k^s = (v^s - \beta_s - z\beta_s)/2$*

When $z = 0$, the total sum remains constant, and the market converges to the benchmark. When $z = 1$, a reduction in β_s induced by self-preferencing results in a corresponding reduction in the total value, which can be viewed as the market scenario where self-preferencing only affects the 3P ancillary service value without influencing the 1P ancillary service.

The alternative definition of self-preferencing does not influence market equilibrium. However, it affects how the 1P service value and market outcome change following self-preferencing. Based on the market equilibrium detailed in Section 2, we first investigate the impact of self-preferencing on social welfare. To do so, we examine how market participation by buyers and sellers changes following the adoption of self-preferencing practices. Based on Definition 2, we calculate the derivative of n_S with respect to β_s :

$$\frac{dn_S}{d\beta_s} = \frac{\beta_s + 2t - 3tz}{12t}.$$

As the denominator is always positive, the sign of the derivative is determined by its numerator. Correspondingly, $dn_S/d\beta_s$ is negative if $\beta_s < -(2 - 3z)t$, and it's positive otherwise.

As for buyer participation, its derivative with respect to β_s equals:

$$\frac{dn_B}{d\beta_s} = \frac{\beta_b}{6t}.$$

Therefore, the sign of $dn_B/d\beta_s$ is positive when $\beta_b > 0$, and is the opposite otherwise. We then check how the monopoly firm's profit changes when β_s increases. Calculating the derivation of the firm

G 's profit with respect to β_s , we derive:

$$\frac{d\pi_G}{d\beta_s} = \frac{(\beta_s + 2t - 3tz)(\beta_s^2 - 9t^2 + 6tv^s + \beta_s t(4 - 6z))}{144t^2}.$$

Since the denominator is always positive, the sign of $\frac{d\pi_G}{d\beta_s}$ is determined by the numerator. Under the assumption that there is positive seller participation, the equation $\beta_s^2 - 9t^2 + 6tv^s + \beta_s t(4 - 6z)$ is always positive. Therefore, $\frac{d\pi_G}{d\beta_s}$ is positive if $\beta_s > -(2 - 3z)t$ and negative otherwise.

Our analysis yields qualitatively similar results to those in the benchmark. First, under the condition that self-preferencing is profitable for the gatekeeper firm, i.e., $\beta_s > -(2 - 3z)t$, self-preferencing is always beneficial to sellers and encourages their entry. Second, self-preferencing practice can either benefit or harm buyers, depending on their preference for 1P versus 3P sellers. Specifically, when buyers dislike 1P sellers and $\beta_b < 0$, self-preferencing makes buyers worse off. Otherwise, it is the opposite.

Compared to the benchmark, self-preferencing can now have an additional effect on the total market value of ancillary services. While this change does not alter the welfare implications when self-preferencing remains profitable for the gatekeeper, it does affect the gatekeeper's incentives. Specifically, the incentive to engage in self-preferencing decreases with z , which captures the extent to which the reduction in 3P service value translates into a net loss in total market value. Importantly, as long as $z < 2/3$ —that is, less than two-thirds of the value lost from reduced 3P service quality translates into an overall market loss—the gatekeeper still has an incentive to self-preference even when its 1P service is of inferior quality compared to 3P competitors.

This insight highlights that gatekeepers in markets where 1P sellers can readily absorb the transactions diverted from 3P sellers have strong incentives to engage in self-preferencing. A prime example is Amazon Marketplace. Buyers on Amazon are typically focused on price and shipping speed. Since most purchases will be fulfilled regardless of whether the shipping is handled by Amazon (1P) or other (3P) logistics providers, self-preferencing has minimal impact on the overall transaction volume. As a result, the negative externality on total market value is small (i.e., z is low), and self-preferencing remains attractive to Amazon.

B.8 Market Size

In the benchmark, we assume that all firms in the adjacent market treat the market size as exogenous when setting their ancillary service prices. In this section, we relax this assumption for the gatekeeper firm and allows it to move first as an incumbent. This corresponds to a scenario where the gatekeeper firm has a sufficiently high market share in the adjacent market, so that its pricing strategy for the 1P service has a non-negligible effect on market size. The timeline of the game is as follows: First, firm G sets its price in the primary market and ancillary service market. Second, fringe firms choose the prices of their own ancillary services. Given the prices in the primary and adjacent markets, buyers and sellers make entry decisions. Last, sellers who join the market purchase the ancillary service that gives them the highest value. Based on sellers' choice of ancillary services, buyers and sellers realize corresponding surpluses in the market. To avoid trivial market equilibria, we make the following assumptions in this section:

Assumption 3. *In Appendix B.8, we assume:*

- I) **[Positive Demand]** *the ancillary service values satisfy $-5t < \beta_s < t$ so that both 1P service and 3P service derive a positive demand at equilibrium; and*
- II) **[Competitive Market and Market Participation]** *the surplus v^s and v_G^b satisfy the conditions such that there exists competition between the 1P service and 3P service, and user entry does not hit corner solution.*

Within this model setup, we proceed to solve for the subgame-perfect equilibrium. The relaxation of the assumption does not affect the transaction stage in the adjacent market. Thus, the demands for ancillary services remain the same as in the benchmark. For the fringe firm k , its profit maximization problem remains unchanged, and its optimal pricing strategy is:

$$p_C = \frac{t + p_G - \beta_s}{2}.$$

As for firm G , it earns additional profit from providing the primary product and derives $\pi_G = m_S n_S + p_G Q_G$. Plugging in the 3P ancillary service prices, we calculate seller entry and obtain:

$$n_S = \frac{\beta_s^2 - 2\beta_s p_G + p_G^2 + 6\beta_s t - 16t m_S - 14t p_G - 7t^2 + 8t v^s}{16t}.$$

Under Assumption 3, each firm derives a positive demand. Plugging in the market size and maximizing its profit, the gatekeeper firm adopts the following pricing strategy at equilibrium:

$$\begin{aligned} p_G^* &= \frac{\beta_s - t}{3}, \\ m_S^* &= \frac{18tv^s + 5t^2 - 4t\beta_s - \beta_s^2}{72t}. \end{aligned}$$

Interestingly, when the gatekeeper endogenizes the impact of 1P service price on market size, it is willing to subsidize sellers in the ancillary service market. Accordingly, the equilibrium demands for market participation are given by:

$$\begin{aligned} n_B^* &= v_G^b - \frac{\beta_b(t - \beta_s)}{6t}, \\ n_S^* &= \frac{\beta_s^2 + 4\beta_s t - 5t^2 + 6tv^s}{24t}, \end{aligned}$$

and the equilibrium profit for the gatekeeper firm equals:

$$\pi_G^* = \frac{(\beta_s^2 + 4\beta_s t - 5t^2 + 6tv^s)^2}{576t^2}.$$

To investigate the impact of self-preferencing on market competition, we calculate the derivatives of n_S with respect to β_s and obtain:

$$\begin{aligned} \frac{dn_S}{d\beta_s} &= \frac{\beta_s + 2t}{12t}, \\ \frac{dn_B}{d\beta_s} &= \frac{\beta_b}{6t}. \end{aligned}$$

The analysis gives rise to the same conclusion as in the benchmark. Specifically, sellers derive a higher surplus following self-preferencing when $\beta_s > -2t$ and a lower surplus otherwise. As for buyers, their surplus increases after self-preferencing when $\beta_b > 0$ and decreases otherwise.

Next, we check how firm G 's profit changes as β_s increases. Calculating the derivative of firm G 's profit with respect to β_s , we derive:

$$\frac{d\pi_G}{d\beta_s} = \frac{(\beta_s + 2t)(\beta_s^2 + 4\beta_s t - 5t^2 + 6tv^s)}{144t^2}.$$

Under Assumption 3, the gatekeeper will find it profitable to engage in self-preferencing when $\beta_s > -2t$. Otherwise, it is the opposite. Combining the results with welfare implication, we derive qualitatively similar conclusions to the benchmark model. When the gatekeeper is motivated to practice self-preferencing, sellers always benefit, while buyers may be either better off or worse off, depending on their preferences for ancillary services.